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Strava Glossary

Relative Effort
Relative Effort is an analysis of your heart rate data. By tracking your heart rate through your workout and its level relative to your maximum heart rate, we attach a value to show exactly how hard you worked. The more time you spend going full gas and the longer your activity, the higher the score. Compare your Relative Effort with friends and pros, see if you can do a truly epic workout and motivate yourself to push that extra bit harder! Relative Effort was inspired by the concept of TRIMP (TRaining IMPulse) coined by Dr. Eric Bannister.

Perceived Exertion
Perceived Exertion lets you manually record how intense your efforts feel on a 1-10 scale ranging from "Easy" to "Max Effort." When tracking how difficult a workout feels overall, Perceived Exertion can stand in for an athlete's heart rate data. That means Perceived Exertion can power features that otherwise require heart rate data, like Relative Effort and Fitness. Add Perceived Exertion to activities for a layer of qualitative data, or when you happen to forget your heart rate monitor.

Energy Output
Energy Output measures the amount of work you've done during a ride, expressed in kilojoules (KJ). It is a factor of how much you're pedaling, how fast you're pedaling and how much force you're exerting on the pedals (measured in W). Power output is most accurately taken from a power meter, but if you don't have a power meter we give a rough approximation through our power estimator.

Average Power
Average power reflects your average power value during a ride, expressed in Watts (a measure of how much energy you are exerting onto the pedals). This is inclusive of the entire ride, and takes coasting into account as well. Average power is most accurately measured with a power meter, though if you don't have a power meter we give a rough approximation through our power estimator.

VAM
VAM measures your Vertical Ascent in Meters/hour – it measures how quickly you are traveling upward. VAM is useful for comparing your effort on different hills and segments, and is used by both cyclists and runners. To get a high VAM score, grades between 6-10% generally present the best opportunity to ascend quickly, as they are steep enough to avoid wind, and gradual enough to allow unrestricted motion.

Intensity
Intensity is our way of showing how difficult a ride was as compared to your FTP. We look at your Weighted Average Power for the ride and compare it to your FTP. If your Weighted Average Power was 250W and your FTP 300W, the Intensity would be 83%. It's very possible to have an Intensity of over 100% if the ride was shorter than an hour.

- Endurance / Recovery Ride – 65% and lower
- Moderate Ride – 65-80%
- Tempo Ride – 80-95%
- Time Trial or Race – 95-105%
- Short Time Trial or Race – 105% and higher

Segments
A segment is Strava's term for a specific section of a road, climb or trail. Strava athletes can create segments from their uploaded activities. Strava tracks your performance on each segment every time you run or ride so you can see how you're progressing over time, and compare your results with other athletes on Strava: your friends, local athletes and even the pros. Start creating segments and see where you land on the leaderboards.

All-time Personal Records
All-time Personal Records (PRs) represent an athlete's fastest time ever run for benchmark distances. PRs are manually entered, and can be linked to official race results and Strava activities.

Whether the PR is from a track, road, or trail race, the PR is often verified by a known time over the race distance. We know that PRs matter down to the second, and we know that GPS data isn't quite as accurate as a time over a verified race course.

How are All-time Personal Records different from Best Efforts?

- All-time personal records are an athlete's best performance ever in an official race or over a verified, known distance or course.
- Best Efforts are training level insights in GPS and power-based activities.

Best Efforts
Best Efforts are calculated using your GPS-based running or cycling activities, and/or cycling activities with power data. Indoor Rides will only count towards power Best Efforts. Strava analyzes the activity to identify the fastest rolling time for benchmark distances, longest activities, activities with most elevation, biggest single climbs and best power outputs at benchmark time intervals. They are considered estimates because they are subject to normal discrepancies in GPS and power meter accuracy.

How are all-time personal records different from Best Efforts?

- All-time personal records are an athlete's best performance ever in an official race or over a verified, known distance or course.
- Best Efforts are training level insights in GPS and power-based activities.

FTP
Your Functional Threshold Power (FTP) is the maximum average power that you can hold for one continuous hour. For example, if you were to ride a 40k time trial in 60 minutes at an average power of 275W, your FTP would be 275W.

FTP is the keystone to training with power. It allows Strava to determine how hard a ride is for you. You doing 300W might feel much different than someone less trained doing 300W and FTP allows us to gauge just how hard segments, rides, and even weeks or months of training were for you!

Calculating Your FTP
We recommend you test for your FTP at least every few weeks to a month while you're training. Here are some tips to get the most out of your FTP testing:

It's extremely taxing on your body (and your training program) to continuously push out 60-minute max efforts. It's also difficult to find a stretch of road where you can ride for 60 minutes uninterrupted and maintain a steady wattage. Thus, the easiest way to calculate your FTP is to test your best average power for 20 minutes. We believe 20 minutes is enough time to stress the same physiological systems as a 60-minute effort would and it is easier to consistently do within your season.

Try to reproduce the same conditions each test – this means use the same stretch of road or always use the same trainer/rollers

Make sure you are fresh (the previous few days should be light in terms of training load)

Properly warm up

Weighted Average Power
When you ride with a power meter, you'll notice how your power jumps all over the place based on the terrain, grade, wind, and other factors. Weighted Average Power looks at all of this variation and provides an average power for your ride that is a better indicator of your effort than simply taking your average power. It is our best guess at your average power if you rode at the exact same wattage the entire ride.

Total Work
Total Work, expressed in kilojoules (kJ), is simply the sum of the watts generated during your ride. There is a close 1–to–1 ratio with Total Work and Calories expended during a ride.

Intensity
Intensity is our way of showing how difficult a ride was as compared to your FTP. We look at your Weighted Average Power for the ride and compare it to your FTP. If your Weighted Average Power was 250W and your FTP 300W, the Intensity would be 83%. It's very possible to have an Intensity of over 100% if the ride was shorter than an hour.

- Endurance / Recovery Ride – 65% and lower
- Moderate Ride – 65-80%
- Tempo Ride – 80-95%
- Time Trial or Race – 95-105%
- Short Time Trial or Race – 105% and higher

Segment Intensity
Intensity on Segments gives you a simple visual indication of how hard you went on a given segment by comparing the effort with your best power for the duration of that segment over the last 6 weeks.

For example, if you maintained 300W for a 10-minute segment and your best 10-minute power over the past 6 weeks was 330W, your segment intensity will be 90%.

Training Load
We calculate Training Load by comparing your power during your ride to your FTP and determining how much load you put on your body during the workout. Training Load is a great way to determine how much rest you need after your workouts. The guide below will tell you how long after a workout it will take you to fully recover:

- About 24 hours – 125 and lower
- 36-48 hours – 125-250
- At least 3 days – 250-400
- At least 5 days – 400+

Power Curve
The Power Curve shows your best average power for time periods of 1 second up to the length of your ride. We search your entire ride and find these best efforts and you can compare them with your best efforts for your last 6 weeks, the current year, or years in the past!

The Power Curve can be displayed in Watts (W) or Watts per Kilogram (W/kg.) on your own activity.

Power Zones
While the Power Curve shows your best efforts for given periods of time, Power Zone charts take each 1 second of power of your ride and put it into a bucket. The buckets are based on your FTP and are as follows:

Active Recovery – Social pace with very little physiological effect on your body. Can be used in between intervals and for recovery rides.

Endurance – Easy pace that you could ride “all day long” Conversation is still possible with little concentration required.

Tempo – Brisk pace that can be maintained for a few hours that requires concentration when riding alone. Breathing in tempo is rhythmic and may become strained at the upper end of this zone.

Threshold – Moderate to hard effort and leg fatigue that can be maintained for up to 1 hour. Conversation is difficult and concentration is required.

VO2Max – Power that is primarily taxing your VO2Max system. Leg fatigue is high and conversation is not possible. VO2Max can be maintained for 3-8 minutes.

Anaerobic – Extremely hard efforts with severe leg fatigue that can be maintained for 30 seconds to 3 minutes.

Neuromuscular – Sprinting power that is taxing your neuromuscular system and can be maintained for 1-20 seconds.

Strava Segments

Rosie

November 1, 2024

Segments are one of Strava's coolest features. They are portions of roads or trails created by members where athletes can compare times. Subscribe to Strava to get the most out of Strava segments.

- Understand why a segment may not have matched your activity and what you can do about it.
- Climbing segments (for cyclists) are automatically categorized HC, 1, 2, 3, or 4 as you see in the Tour de France. More information on how we categorize climbs.
- See Optimizing Segment Creation for what kind of segments work the best and which don't.
- With a Strava subscription, you can use segments to compare your own times or to compare with other users' times who have also completed the segment. Segments are a great way to keep track of your performance progress and see how your training compares to others.
- To learn how to customize your segment viewing preferences on your activities, see Hide/Unhide Segments.
- To find segments in your area or in a new area, see How to Find Segments.
- If you want to be the King of the Mountain (KOM), Queen of the Mountain (QOM), or Course Record (CR) holder, you have to have the fastest time on a segment.
- For a glossary on Segment Achievement medals and how you get them, see Segment Achievement Awards.

My Segments

There are three categories under "My Segments" - my KOMs/QOMs/CRs, which lists those segments in which you are currently first place, then Segment Preferences, which keeps track of what segments you have hidden (essentially deleted them from your viewing experience but not for other users) or unhidden (chosen to view that others have hidden) then there are Created Segments, those you have personally created and which you "own" effectively.

What happens to my segments if I delete my account?

Any private Segments you created will be deleted when you delete your account. Any public segments or routes you've created, including the current title, will remain on the Strava platform, allowing the community to continue to record and compete on them.

- As always, we encourage you to refrain from using personally identifiable information in your titles.
- Your original activity and GPS data will no longer be linked to the segment, and your activities will no longer appear on the segment leaderboards.

If you would like to remove a public Segment from Strava regardless of the above, you can delete the segment before deleting your account. There is no way to make a public segment private.

Changes to Segments

We don't typically make changes to segments. However, in order to ensure Strava segments are the most useful for the Strava community, we may edit, adjust, make private, or delete segments at any time. Learn more about our segment cleanup efforts here.

Verified Segments

Strava also occasionally "verifies" top segments so they can be best seen and used by the community. Learn more about Verified Segments here.

What’s a segment?

Meg

April 3, 2023

Your GPS-based activities can be broken down into smaller sections called segments. Segments designate specific features or portions of a route – such as a climb, a tricky stretch of trail, or part of an open-water swim. Each time you complete a segment, your time is recorded. With a Strava subscription, you can compare it to previous efforts and those of your friends and other athletes.

Each activity type has its segments and leaderboards. Cyclists compete with cyclists, runners compete with runners, swimmers with swimmers, hikers with hikers, etc.

You’ll find that many of your activities contain pre-existing segments, but you can also create your own using the Create Segment tool found on your activity pages on the web.

Earning achievements

When you upload an activity to Strava as a subscriber, you may be awarded “achievements” for outstanding performances on the traverse segments. The different types of achievements are listed below.

Crown — King/Queen of the Mountain (KOM/QOM) or Course Record (CR)

The crown signifies the top performer in a given segment. Crowns are awarded to the fastest athlete to date on a segment’s overall leaderboard.
Your current all-time KOM/QOM/CR crowns are stored in a special list on Strava for your reference. Access the list under "KOMs/CRs/Top 10s" or "QOMs/CRs/Top 10s" from your profile page on the web or under “Segments” from your profile page on the mobile app.

Trophy — Overall Top 10 Placement

A trophy is awarded when you place near the top of an overall segment leaderboard. Trophies exist for 2nd place to 10th place all-time efforts.
Your current top 10 placements are stored in a special list on Strava for your reference. Access the list under "KOMs/CRs/Tops 10s" or "QOMs/CRs/Top 10s" from your profile page on the web or under “Segments” from your profile page on the mobile app.

Medal — Personal Top 3 Placement

Medals are awarded for your best personal performances on a segment. If you clock a new personal record on a segment, you'll receive a "PR" medal. Likewise, for your second and third fastest times.
For runners, medals are also awarded for your best performances over the following distances: 400m, 1/2 mile, 1k, 1 mile, 5k, 10k, half marathon, and marathon.

Laurel Crown — Local Legend

The laurel crown indicates that you are the Local Legend of a segment. To earn the Local Legend achievement, you must match to the segment more times than any other athlete in the past 90 days.

Create a Segment

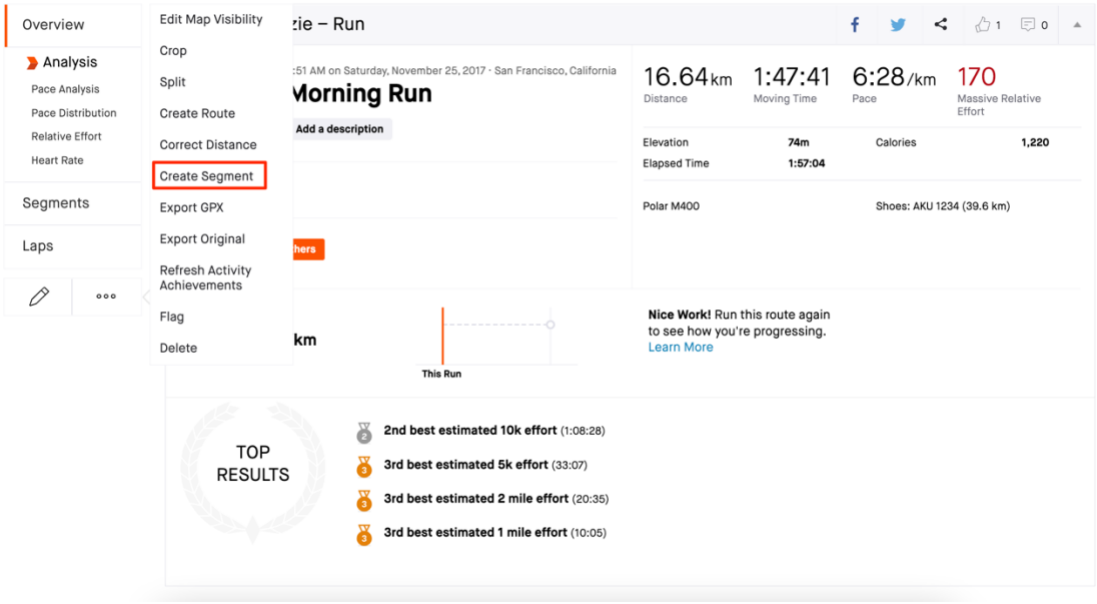
Rosie

January 29, 2025

https://support.strava.com/hc/en-us/articles/216918157-Create-a-Segment

Strava users can create segments to mark the most interesting parts of their activity routes. Segments need to have a start point, an endpoint, and a sequence of locations in between. The first key to segment creation is that the data has to already exist within one of your own Strava activities. Each segment will be based on the activity data from which it was initially created.

- Begin by viewing one of your existing activities on the Strava website. **Please note that segments cannot be created on the Strava mobile app**. Whatever activity you choose, that data will be the foundation of the segment and affect the segment's accuracy and matching.
- Click on the more (ellipses) icon and then click on **Create Segment**.



- You'll then see the segment creation interface, where you can specify the start and end using the green and red sliders above the map.
  - As you drag the sliders forward and back, the map updates the position of the green start marker and the red end marker.
  - For **single-step** adjustments, you can use the back and forward buttons below the slider's left and right ends.
  - Here are some helpful tips for optimizing segment creation.
- When you are happy with the start and endpoints, click **Next**. Strava will check to see if a similar segment already exists.
  - If Strava identifies that a duplicate segment of your proposed new segment exists, you can see it on the map and compare it to your proposed segment. If you click on the name of the duplicate segment, you'll be able to view the leaderboard tab. You can create a duplicate of an existing segment, but it must be private only to you.
  - If Strava identifies that a similar segment(s) exists to the one you're trying to create, you'll be able to see it on the map and compare it to your proposed new segment. If you click on the name of a similar segment, you'll be able to view the leaderboard tab. If, after inspecting the similar segments, you believe your proposed segment is different from those similar segments, you may

create a new public segment. Please first double-check that your proposed new segment does not already exist.
    - If prompted, click **Next** again, and you will have the opportunity to name the segment. Good names describe the segment in a way that others will recognize. It can be helpful to include the name of the landmarks used for the start and end or use a familiar nickname known to locals.
- The final step will be to decide if the segment is private or public. Private means that you will only see the segment and only your results will be shown. Your results on the segment will not be visible to others when they view your activities that match the segment. Public means that the segment will be visible to all users.
- Once you click **"Create my Segment,"** you will be returned to the activity page where you began. You will be able to find that segment on this activity and in your **"Created Segments"** tab on your **"My Segments"** page. It may take several hours for a historical leaderboard to be created, so check back soon for complete results.

Optimizing Segment Creation: How to Create Good Segments

Rosie

January 29, 2025

https://support.strava.com/hc/en-us/articles/216918227-Optimizing-Segment-Creation-How-to-Create-Good-Segments

Certain rules follow when creating segments that can minimize false positives, missing segments, inaccurate matches, and more.

- Very Short Segments
- Bad Start/End Points
- Tight Switchbacks
- Parallel Trails
- Very Long Segments
- Very Complex Segments
- Tunnels
- Segment Creation Tips

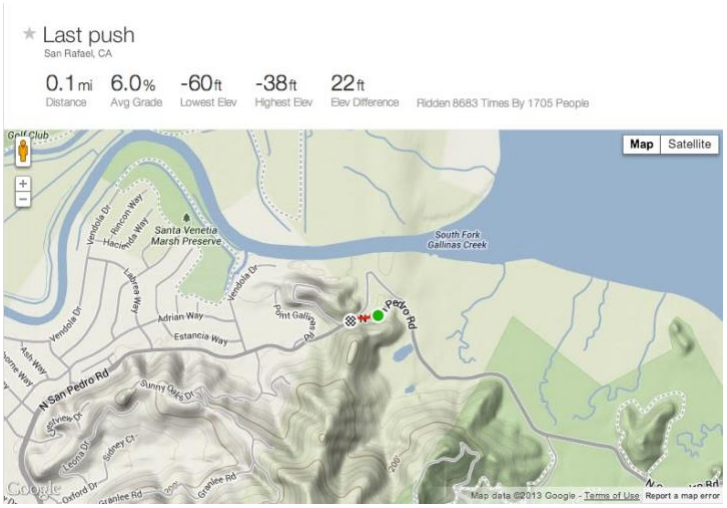
Getting Started

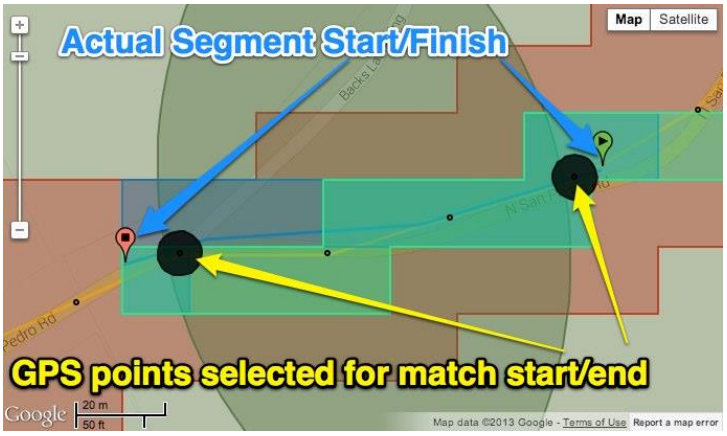
Zoom in! The closer you can see the data that you're working with, the better the end result will be. Switch away from Terrain Maps (at the top right of your map display) to either Standard or Satellite maps to enable you to zoom in further. Keep in mind that when you're at the segment creation page, the "Move Start/End Point" buttons are dynamic; when you are zoomed out, they will move the endpoints of your segment in large increments - as you zoom in, they will make finer adjustments.

Very short Segments

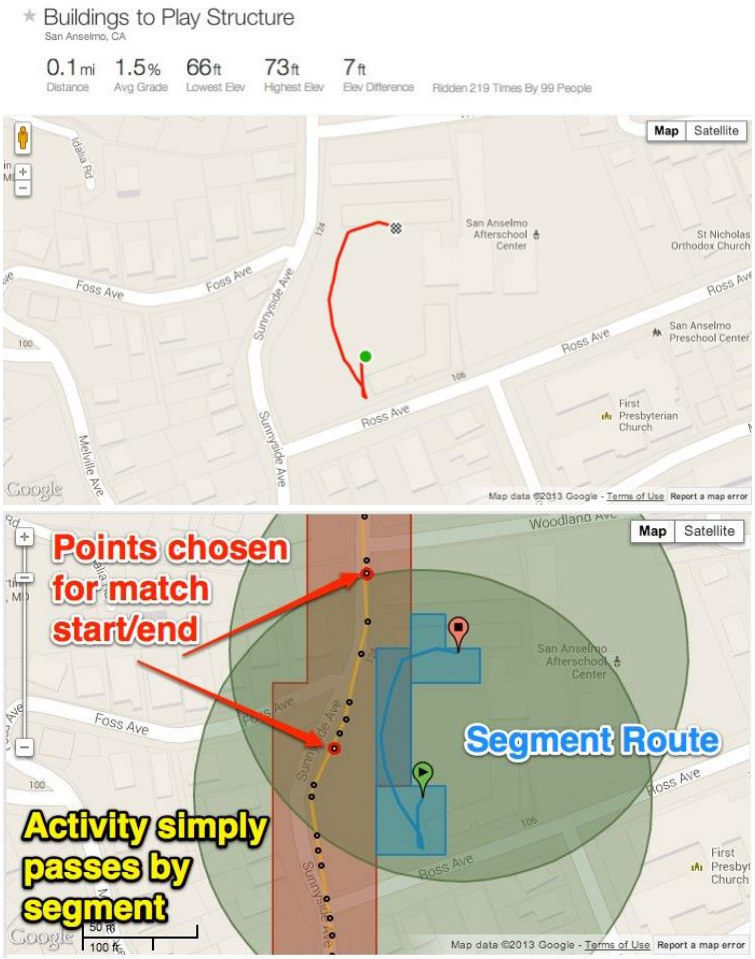
Very short segments - for example, sprints that take less than 30 seconds to complete - will be much more prone to matching errors than longer segments. There are two severely limiting factors for segments like this.

First is the sampling rate of your GPS device; some devices only record a GPS point every 5 to 10 seconds - meaning that the margin of error of your segment match accounts for a very high percentage of the total segment time since the segment is so short. In the example below, you can see that no GPS point falls exactly on the start and end-points for the segment and that the entire segment match is comprised of only four GPS points.



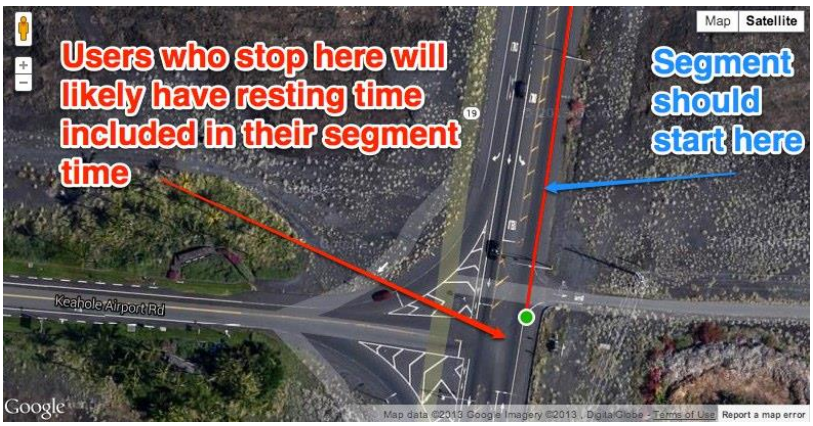


Furthermore, since the segment is so short - in terms of distance this time - the start and end-points will be very close together. There is a degree of leniency necessary in our segment matching system to account for inaccuracies in GPS data, and it's possible that both the start and end-points of a very short segment can fall within the 'buffer' zone intended to allow for a slight GPS drift. This can cause our system to match the segment when there is hardly any overlap between the segment and the ride.



Segments with bad start or end-points

Segments created with start points in less than ideal positions can also be very problematic. For example, if you create your segment in such a way that it starts too early, some users may have trouble matching that segment; see below left for an illustrated example of this.

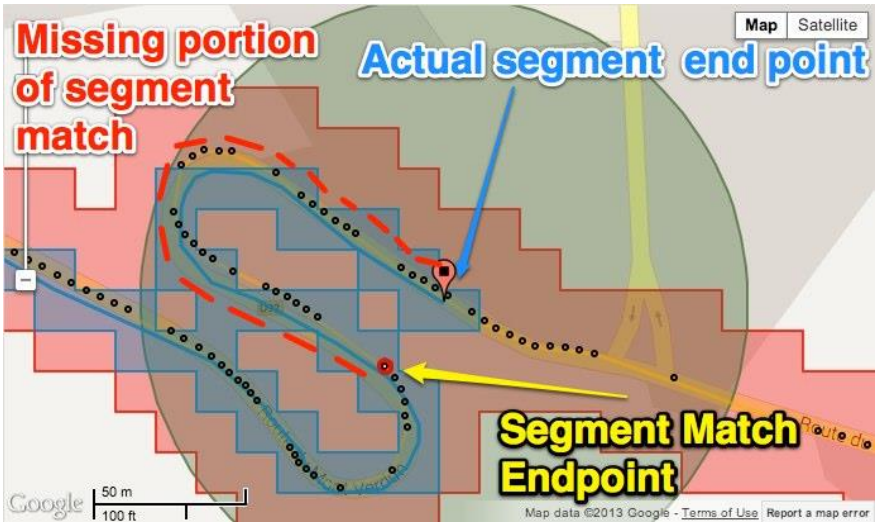


Additionally, placing your segment start point too close to an intersection might lead to inaccurate segment times. The reason for this is that if there is a stop sign or other traffic regulation for the intersection at which the segment starts, users might trigger the start of their segment match before stopping - meaning that their time, while stopped at the intersection, will be included in their time for the segment. See the above right for an example of this.

The reverse of both of these examples is also true; placing the end-point of your segment too close to an intersection can lead to stopped time being included in a segment effort. This is very common at the top of climbs; when a segment finishes too close to an intersection, users will often stop to rest after pushing themselves up the climb - causing their resting time to be counted into their segment time. Likewise, placing your segment start point after an intersection can cause some users not to match at all, if they don't turn in the same direction that the segment does.

Tight Switchbacks

Segments that start or end on a tight switchback turn can cause bad matches. Because our system allows for some small margin of error to account for GPS drift, it is possible that you can trigger the start or end of a segment before or after you actually should have - if the road or trail makes a very tight turn and turns away from, then returns close to the true endpoint. We are hard at work on improving this situation and hope to handle these segments better in the future. In the meantime, the best way to increase your chances of a good match to segments like the one shown below is to extend the segment as far from its own 'tail' as possible (while conforming to the other guidelines discussed on this page).



Trails that parallel roads

Because our system allows for some small margin of error to account for GPS drift if a trail segment runs parallel to a road segment and the two are close enough together, it may not be possible for Strava to provide mutually exclusive results. This means that athletes riding or running on the road may match (and receive unrealistic times) on the trail segment, and vice versa. Unfortunately, if the two paths are close enough together for this to be an issue, the only way to improve your chances for accurate matches is to move the segment endpoints further away.

Very long segments

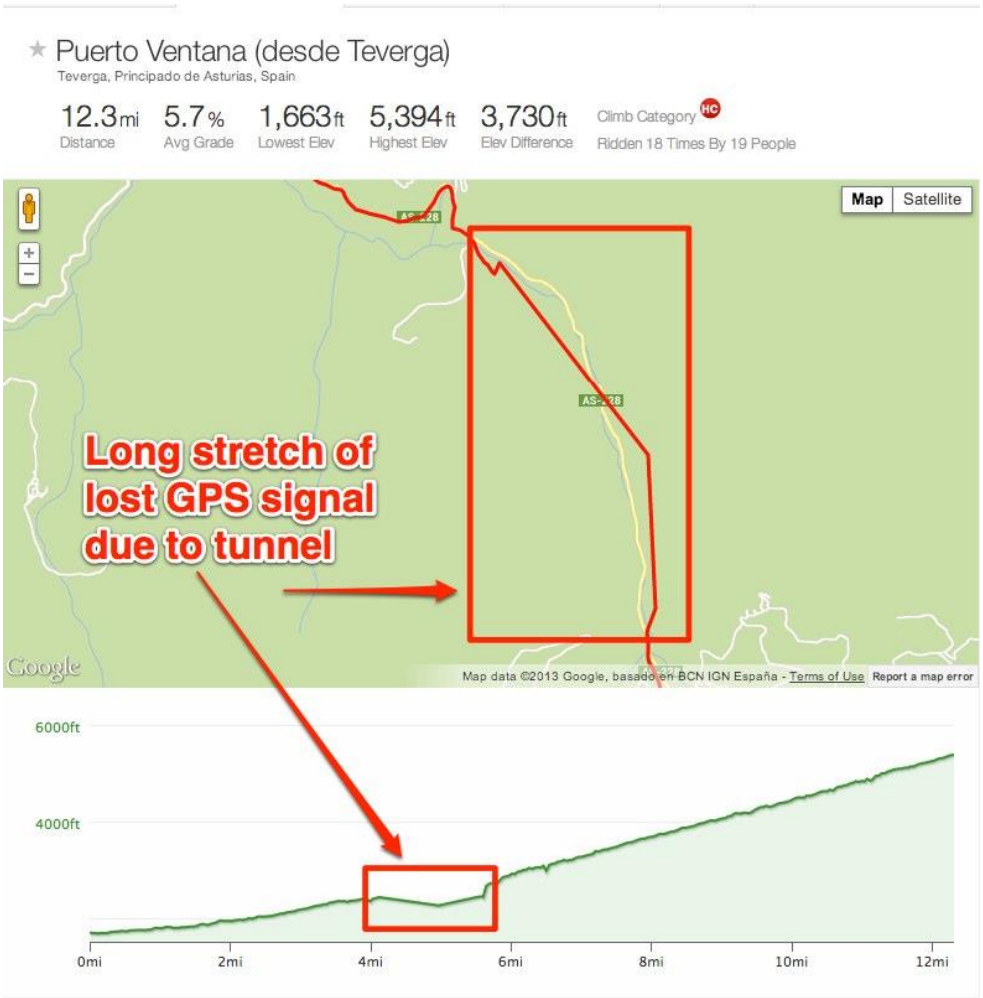
Longer segments are generally more accurate than shorter ones - however only to a certain extent. If you are completing a 30-mile-long segment but make even a very small deviation from the route that the segment follows, you won't be able to match the segment. Even though a one-block discrepancy might have little effect on your time for the segment, any deviation from the segment's route will prevent it from matching your activity. See below for an example of this.

Very complex segments

Certain segments are just so complex - composed of so many tight twists and turns - that we are unable to give completely accurate results. This is sometimes evident on cyclocross courses, tightly-packed trail networks, etc. This is exacerbated by the fact that segments like this are most often found off-road - with tree cover frequently negatively affecting the quality of both the segment and the activity GPS signal. Once again, there, unfortunately, isn't a real solution to this problem other than to use the best available GPS data, and whenever possible follow the other guidelines described here.

Tunnels

Any time your route takes you through a tunnel of any significant length - as is common in mountainous areas - your GPS signal will be lost, and the distance you travel while without a GPS signal will likely be long enough to trigger our 'Gap Threshold.' This means that segments that incorporate tunnels will be unlikely to produce accurate matches to your activities. Unfortunately, there isn't a great workaround for this - the best solution is to simply avoid creating segments that include tunnels longer than 500 meters.



Other Things to Keep in Mind

Use good quality data!

Quality GPS data is key to accurate segment recognition. Since we use the GPS data from your activity to create segments, it's crucial that segments are created from activities with high-quality GPS data - and that, whenever possible, are recorded on a high-quality GPS device. Not all GPS devices are created equal but even for the most accurate devices, trees, tall buildings, and even clouds can wreak havoc on GPS signals. Look closely at your activity before choosing it as the basis from which you create your segment; if the GPS data looks jumpy or doesn't follow your route closely, try using another activity.

Timing Issues

Recording intervals vary between devices - for example, the Strava mobile app records every second while Garmin devices use either 1-second intervals or a smart recording which has a varied recording interval. Segment matching works the same on each GPS dataset, but depending on the device's recording interval, can yield different results. Segment matching uses the GPS points in the data closest to the start

and endpoints of the segment, and as this can vary with each activity, timing on a segment can vary slightly because of this. At the present time, we don't interpolate or extrapolate GPS data to normalize the exact start and end positions of the segment.

Are there duplicates?

Check to see if the segment already exists. After you specify a start and end-point and click "Next," you may see a duplicate segment or a list of similar segments. If the segment is a duplicate of a segment that already exists, you will only be able to create a new private segment. If Strava identifies similar segments, review them to see if the new segment you're proposing is truly new. For more information on Creating Segments, see here.

Re-Routed off-road trails

If you are attempting to create a segment to account for an off-road trail that has been re-routed in response to environmental changes, there, unfortunately, isn't a way to prevent users on the old 'version' of the trail from matching the new one - or vice versa - assuming the two trails are similar enough to match regardless of which route was followed. The best option is to filter segment results by 'This Month,' 'This Year,' etc.

Segment Matching Issues

Rosie

December 28, 2023

Understand why a segment may not have matched your activity and what you can do about it, why false matches occur, and common problems in our segment-matching algorithm.

- How Segment Matching Works
- Missing Segments
- Map Visibility and Segment Matching
- Incorrect Matches
- Problematic Segments

How Segment Matching Works

It's important to keep in mind that each GPS device records its own unique set of data even if, for example, you and your friend are recording the same activity on the same day. When we match you to a segment, we choose the GPS points from your file that fall closest to the segment's start and end. Because of the nature of GPS and the differences in device recording intervals, it's unlikely that there will be a GPS point exactly on top of the segment's start/end. This means that you can be timed over portions of your activity that are slightly longer or shorter than the actual segment length. Everyone is susceptible to this, but you may get a short match that works in your favor or a long match that adds a few extra seconds.

What can I do? The more GPS data points you record, the more options Strava has when choosing a start and endpoint, and the more precise your match is likely to be. Your device's recording interval determines how often you record GPS points. Recording intervals often differ from device to device, but if you have the option to set your recording interval, we recommend one-second intervals.

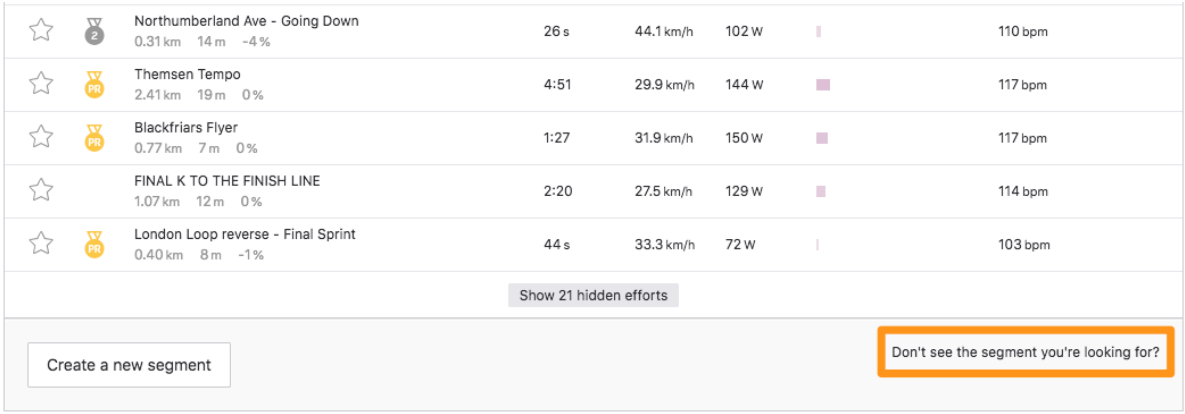
My Activity did not match a segment that it should have

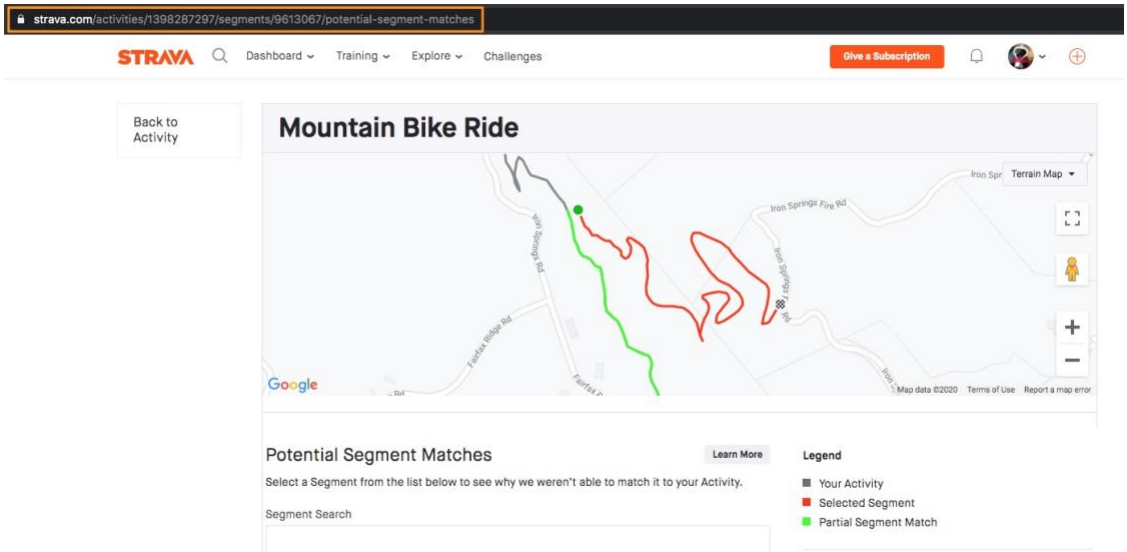
Why does this happen?

- The most common reason a segment doesn't match on a ride or run is that GPS drift occurred while you were recording your activity. This can cause you to miss the start or endpoints of the segment - both of which are necessities - or can cause you to trigger our "Gap Threshold" in the middle of a segment.
- One other possibility is that it is "hidden." There is a link at the bottom of the ride page (website only) that may say "Show hidden Segments." You may see the segment listed there - and you can unhide it. For more details, see the article here.

What can I do?

- If you are looking for details about why a specific segment might not have matched your activity, visit the activity on the Strava website and scroll down to the bottom of the list of segments. Click the link 'Don't see the segment you're looking for?' (highlighted below) to open the Potential Segment Match Analysis tool.
- **If a segment is missing from your activity when it should have matched, please proceed with creating a support ticket and using the Potential Segment Match tool to generate a link.** In the analysis tool, select the missing segment and copy the link in the address bar of the webpage. Paste the link into the support ticket and repeat this process if there are multiple missing segments.
- Alternatively, you can send us a link to the activity page and the link to the missing segment.





Map Visibility and Segment Matching

The map visibility settings applied to one of your activities can affect the segments that match that activity. When a segment starts or ends within a portion of your activity that is hidden, that segment can still match your activity, however, the segment match will be a private match on your activity. Private segment matches are only visible when you view your activity. No one else on Strava will be able to see the private segment matches and they are not eligible to appear on public segment leaderboards or receive public achievements.

If you would like the private segment match on your activity to become a public match so that it is eligible for the segment's public leaderboard and achievements. In that case, you will need to adjust the map visibility settings on the activity to ensure that the segment in question does not start or end within a hidden portion of your activity. Once your changes are saved, the segment will be eligible for the public leaderboard.

Please note that the map visibility settings on the Settings page of your Strava account are different than the Map Visibility settings that exist on an individual activity. Excluding changes to a hidden address, changing the Map Visibility settings on your Strava account will not affect previously uploaded activities. Instead, these settings are only applied to new activities uploaded to your Strava account. To change the Map Visibility settings on an individual activity, you will want to navigate to that activity on your account and use the ellipsis button (...) to find the “Edit Map Visibility” option. From there you will be able to change these settings for the activity you are viewing.

My activity matched a segment that should not have

Why does this happen?
Occasionally a ride or run will be falsely matched to a segment or receive a wrong segment time. Sometimes this is due to the fact that our segment-matching algorithm must be flexible enough to account for GPS drift and can be a little loose with the matching. Poor GPS data can also contribute to inaccurate segment times. There may also be instances where you received a lift or rode in a vehicle for a portion of your activity, and the Crop tool can remove that section of your activity and the invalid segment times.

What can I do?
If you'd like to be unmatched, please proceed with creating a support ticket and include a link to your ride or segment effort, and we can investigate. We can only unmatch segments in cases where it's unlikely there will be future problems with that activity data.

Problematic Segments

There are specific rules to follow when creating segments that can minimize the number of problems that segments will experience - including false positives, missing matches, inaccurate matches, etc. We highly suggest avoiding the start or end of a segment if you are going to stop or turn around, as the resting time may be included in your segment effort. Here is some more information on Optimizing Segment Creation.

Using the Potential Segment Match Analysis Tool

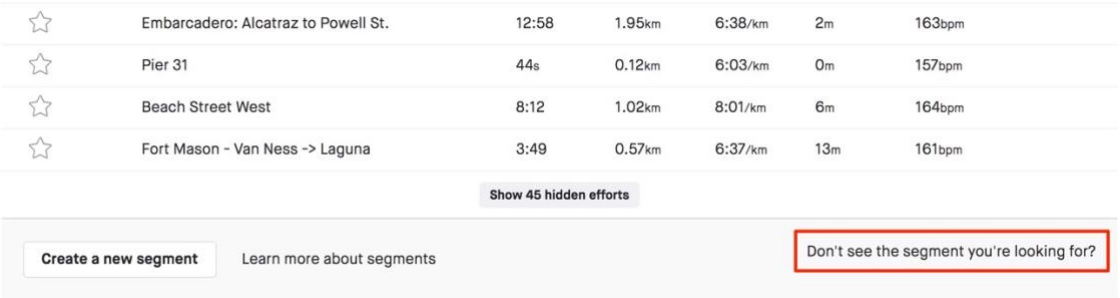
Meg

August 15, 2023

How to find and analyze missing segments:

To learn about why a certain segment didn't match, visit the activity on the Strava website and scroll down to the bottom of the list of segments. Click the link '**Don't see the segment you're looking for?**' to open the **Potential Segment Match Analysis tool**.

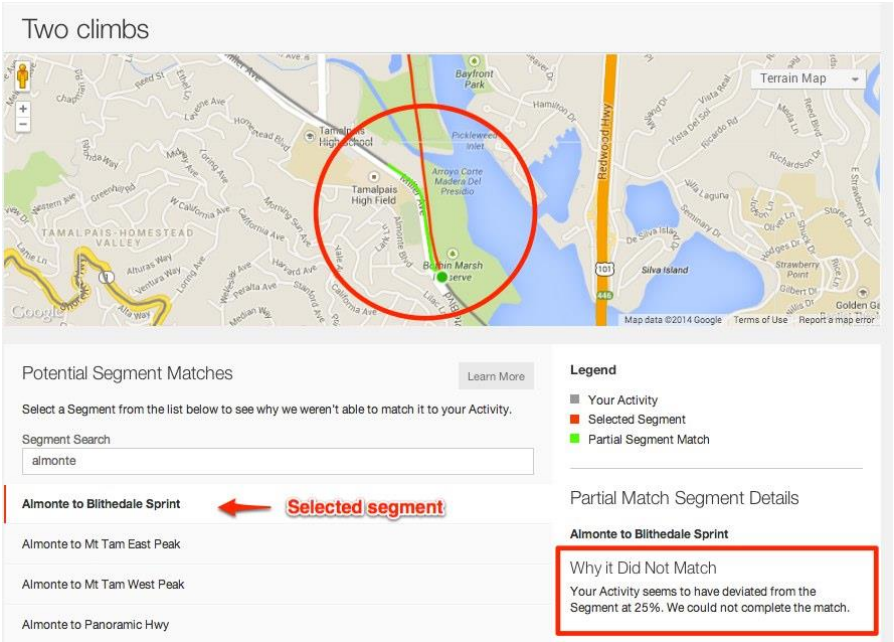
Looking at this tool, you'll see a map of your activity, as well as a list of segments that have a partial or potential match to your activity. You can also search this list by segment name.



What causes a missing segment?

Use the Potential Segment Match Analysis Tool to gain insight into what prevented the segment match and where: GPS errors, missing data, or GPS drifting during recording are the most common reasons a segment doesn't match an activity. For example, poor or missing GPS data around the segment start or end may mean that there is no accurate activity data to create the segment match.

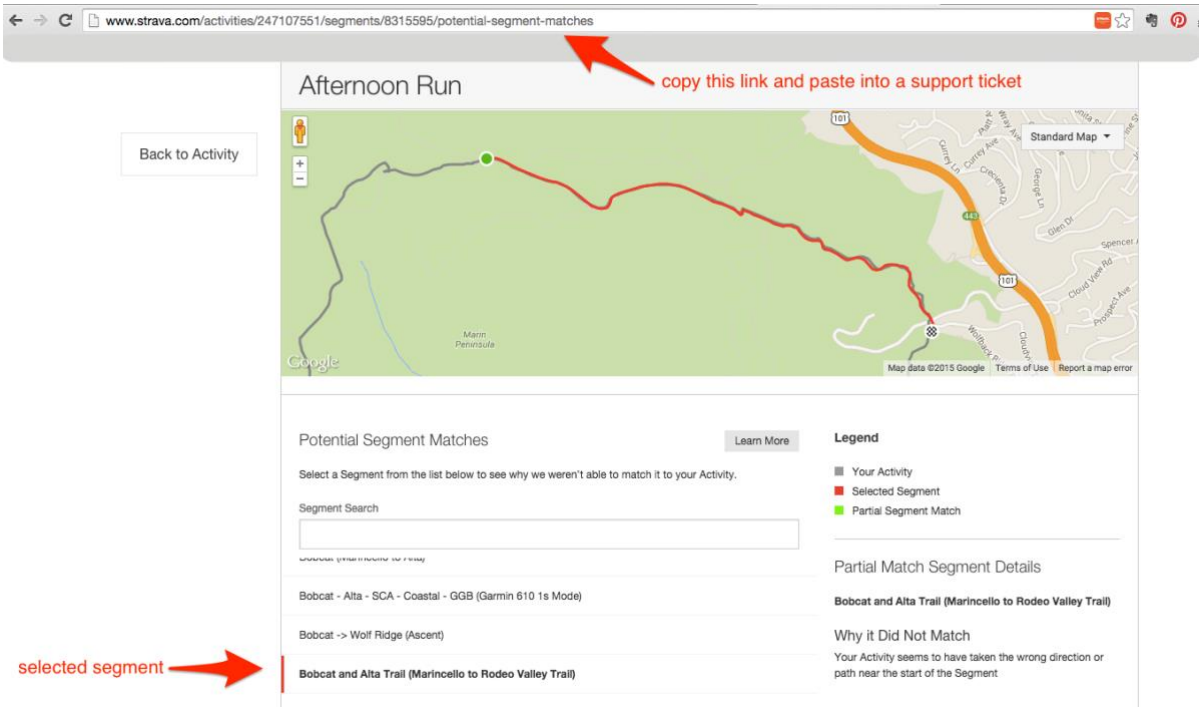
In the Potential Segment Match Analysis tool, compare the grey line which represents the path of your activity to the red line which represents the path of the segment. The portion of your activity that 'matches' the segment will be highlighted in green - and you'll be given some brief information about what might have prevented the full match.



How to request a manual match for a missing segment:

If a segment should have matched your activity but didn't, send us the link to the **Potential Segment Match Analysis tool** for that segment. From your activity page on the website, scroll down and select the option '**Don't see the segment you're looking for?**' below the list of existing segments.

In the analysis tool, select the segment that's missing and copy the link in the address bar of the webpage. This link contains information about your activity and the missing segment. Paste the link into a support ticket and repeat this process if there are multiple missing segments. Sometimes we can manually create the match but in some cases, we can't.



Resting Time in Segment Efforts

Meg

December 28, 2023

https://support.strava.com/hc/en-us/articles/115000248810-Resting-Time-in-Segment-Efforts

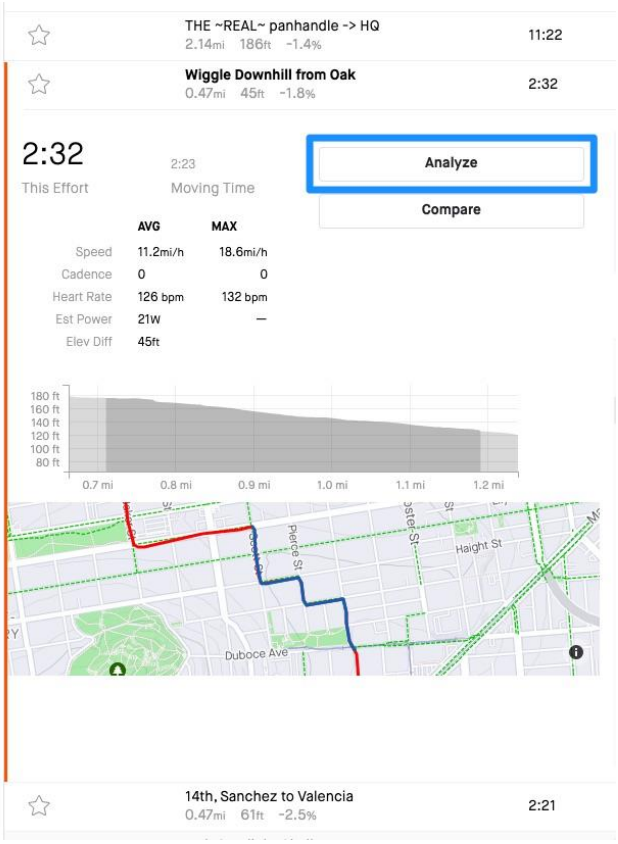
All our segments are ranked according to 'Total Elapsed Time.' We like to think of our segments as a race, and in a race, the clock keeps running even if you stop. Resting time can be captured in a segment effort if you rest or turn around near the beginning or end of the segment.

- Identifying resting or turnaround time
- Why does this happen?
- Preventing the issue

Identifying resting or turnaround time

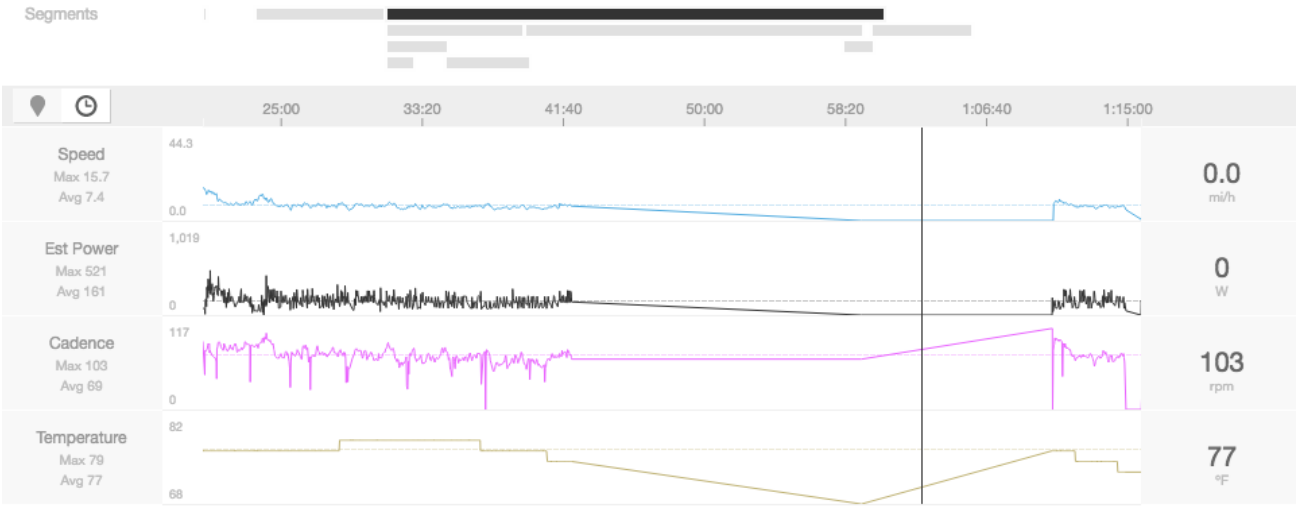
With a Strava subscription, you can view your segment efforts on the Strava website. Depending on the activity type, you may be able to analyze your segment efforts and identify resting or turnaround time.

- Certain activity types (including rides and snowsports) will allow you to click on a segment from the list and select **Analyze** (highlighted below) from the expanded segment view.

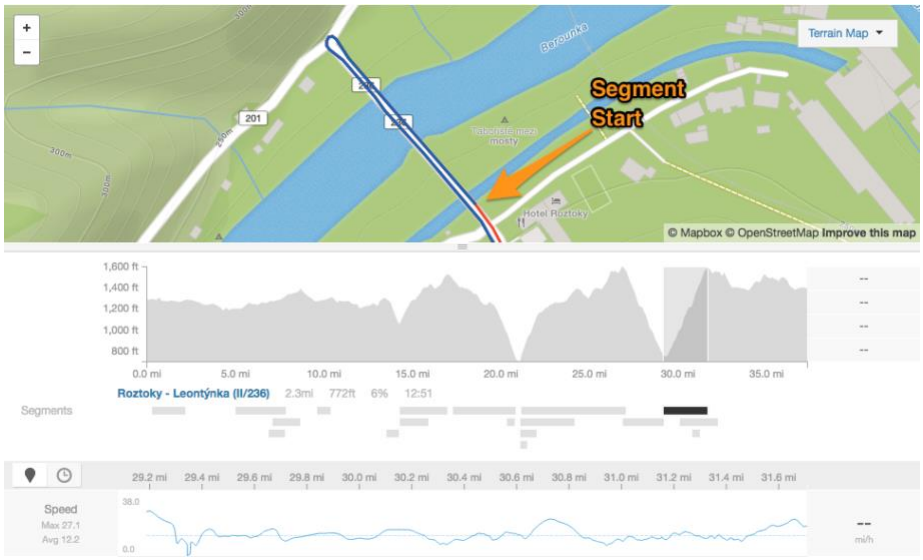


- You can compare your speed on the analysis graph with your location on the map to identify where you were going and at what speed. Select the clock icon (highlighted below) to view your speed charted over time

- If there is resting time included in your segment effort, you will see your speed flatline (likely around 0 miles or km per hour, as shown below) for an extended period.



- Turnaround time is a bit harder to identify because you are still moving. You can look at the map to see what chunk of your ride was timed for that segment (highlighted in blue.) In the example below, the path time for the segment effort is highlighted in blue, and you can see where the athlete turned around before the segment started.

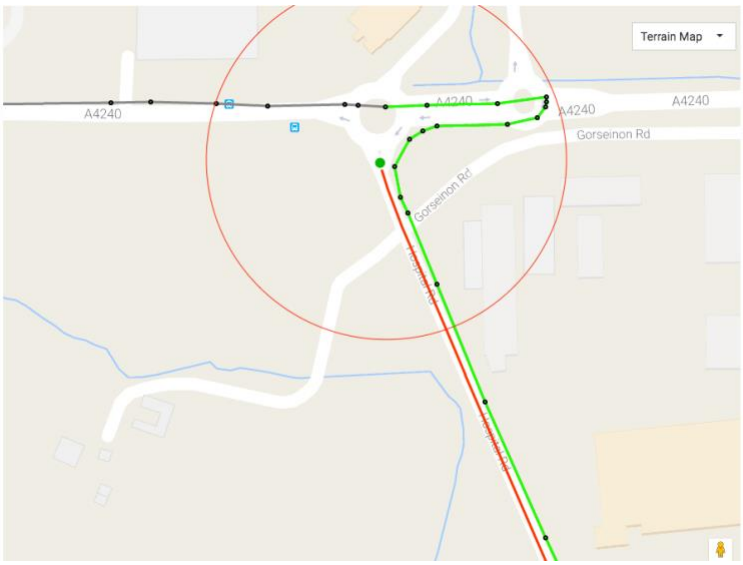


If your activity type does not have the option to analyze segment efforts (running and hiking, for example), it may be more challenging to identify whether there is resting or turnaround time is included in your segment effort. However, these segments are still ranked according to elapsed time, and we still recommend taking steps to prevent this issue.

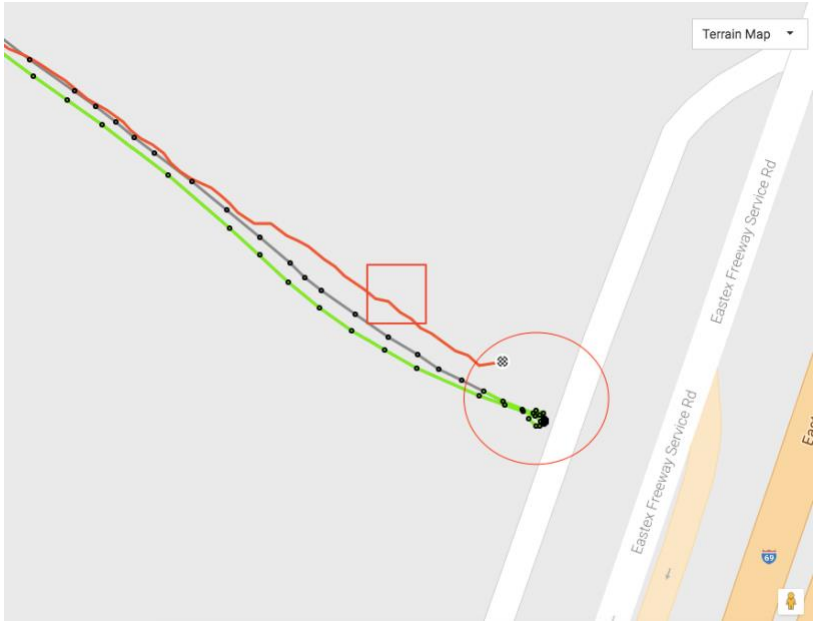
Why is resting time sometimes included in segment efforts?

As you've probably noticed, GPS data isn't perfect, and there is often some margin of error with your data. To account for moderate GPS drift, our segment-matching algorithm has to be a little loose. If our matching process was more strict, you'd often miss out on segments.

- If you cross through the segment start point range to stop or turn around, you may prematurely trigger the segment timer. In the example below, moving past the segment start point triggered the timer and the athlete did not travel far enough beyond the start to re-trigger the segment timer when they returned.



- If you stop or turn around near the end of the segment, you risk not moving far enough beyond the segment endpoint to trigger the segment timer to stop. In the example below, the athlete stopped after just finishing the segment, and their resting time was counted towards the segment effort.



Preventing resting time in segment efforts

Unfortunately, in this scenario, we're unable to adjust your segment effort time. However, there are a few ways you can prevent issues in the future:

- We always recommend riding *through* a segment's start or endpoint. If you need to turn around or stop, we recommend avoiding the start or endpoint by about 70 meters.
- It may not be possible to avoid this issue if the segment has a bad start or endpoint. We recommend placing the end of the segment just before the natural stopping point and the segment start point just past the actual starting point for the best results. Learn more about Optimizing Segment Creation.

Strava Verified Segments

Rochelle Garcia

September 17, 2024

What are Verified Segments?

Strava is full of segments created by our users. **Verified segments** help highlight the iconic stretches of road and trail where the community competes. Strava inspects segments from our segments database, identifies which are the most representative of the road or trail, and verifies them. This feature aims to reduce confusion and improve the accuracy of the world's most popular segments.

How do we Pick Verified Segments?

Verified segments are chosen based on their popularity on Strava. We use a blend of matched activities, leaderboard views, goals, “stars,” and other technical factors related to the segment geometry. Whenever possible, we work with local experts and event organizers to ensure verified segments' start and end points are in the correct location. We also inspect every segment section to ensure it follows the proper track. Segments are based on GPS data from activities, so we're looking for segments that accurately match the underlying road or trail.

What happens when we verify a segment?

When we identify a segment for verification, we take a few steps to ensure it is most useful for the community in the future.

1. **We use our elevation dataset to ensure accuracy.** Data from uploaded activities also increases the accuracy of the segment's elevation data, helping everyone get a correct picture of the segment's elevation profile.
2. **We align the segment to the basemap to ensure accuracy.** This helps ensure that activities that follow the segment's track match the segment.
3. **We move the segment to a Strava Admin account to ensure it isn't edited or deleted.**

If a segment you created becomes verified, it will no longer appear in the "**Created Segments**" tab of your **My Segments** page on the Strava website. Strava Admin becomes the owner and administrator of the segment to preserve it for future use by the Strava community. If you notice this has happened with a segment you created, we sincerely appreciate you making a segment for the community!

Strava Live Segments on ActiveLook

Meg

December 28, 2023

Requirements

- Strava subscription
- ActiveLook Apple Watch app

Linking Your Accounts

1. Open the ActiveLook app.
2. Tap the setting icon (bottom right corner) > “Sensors, watches, Apps and more.”
3. Tap “Strava” under Connected Apps on the next screen.
4. You will be prompted to log into your Strava account or create an account if you do not have one. Accept that the ActiveLook app can connect to your Strava account.

After connecting, your ActiveLook activities (recorded from the mobile or Apple Watch app) will be automatically uploaded to your Strava account. You can even sync activities from ActiveLook that were recorded before connecting with Strava.

Selecting Segments

From the ActiveLook Apple Watch app press the settings menu (...) for your selected sport type (running or cycling.) In the “Strava Live Segments” section you can:

- Enable/disable starred segments.
- Enable/disable segments around you. The list of segments “around me” consists of the most popular segments within 2 to 20 km.
- Select “Browse” to view your list of segments. The segments displayed will be sorted by distance to your current location, closest first.

At least one of the “Starred” or “Around Me” toggles must be enabled to view some segments. After the initial connection, you may need to wait a few minutes for the segments to be synced to your watch. Ensure your watch has a stable internet connection to reach Strava services.
Click on a segment card to see the details for this segment.

- Star or unstar the segment by tapping the star icon.
- Control whether individual activities are eligible for live segments by activating or deactivating them. When a segment is de-activated it will still be available in the list of segments but no race will be triggered during your activity.

Please be aware that we do not sync downhill ride segments to devices for Strava Live Segments. Please keep in mind that our system recognizes even the slightest negative grade so there may be segments that display a 0% grade that is considered downhill and therefore will not sync to devices.

On ActiveLook

Choose the opponent(s) you want to race against during your live segment efforts. During the segment effort, a dedicated racing screen will be created for each of the selected opponents if *the opponent has a record recorded*. As a consequence, for example, if you have disabled the KOM as an opponent, and if you do not have any PR, Wolf, or Carrot effort recorded for a given segment, then no race will trigger for that segment. You can swipe through the various racing screens to monitor your performance against each of the available opponents when the segment is active.

1. From the ActiveLook Apple Watch app press the settings menu (...) for your selected sport type (running or cycling.)
2. In the “Strava Live Segment” section select “Race Against.”
3. Enable/disable toggles for the various available opponents:
    - **Your PR:** Your personal record for the segment.

- **The KOM/QOM/CR:** Record holder for the segment.
- **The Carrot:** The racer just ahead of you on the leaderboard who is part of the list of athletes you are following on Strava.
- **The Wolf:** The racer just behind you on the leaderboard who is part of the list of athletes you are following on Strava.

When an active segment is detected in your proximity you will get notified by an “Approaching screen.” This screen will be displayed on your watch and eyewear with key segment information and an indication of the remaining distance to the segment start flag. The Live Segment will only trigger if you cross the segment start line in the proper direction. If you do not cross the segment start line or go more than 150 meters away from the segment start flag then the segment will be automatically discarded and screens will switch back to your previously displayed data screens. Click the “Ignore” button on your watch to dismiss the race for the segment.

If you are racing on a given segment, and approaching another segment start flag, it will be detected and eventually start in parallel to the first one. New racing screens will be added in the background in order not to disturb your ongoing segment. You can swipe to new segment screens on your watch or with the eyewear gesture sensor (if enabled.)

FAQ

How do I disable Strava Live Segments on ActiveLook?

From the ActiveLook Apple Watch app press the settings menu (...) for your selected sport type (running or cycling.) In the “Strava Live Segment” section, toggle **both** “Starred” and “Around Me” OFF.

Why is my segment time different after I upload than it was on my ActiveLook?

ActiveLook provides an estimate of your live position during the segment and finish time at the end of the segment. To ensure consistency between users, the times shown on ActiveLook during an activity are preliminary, and the times shown on Strava after uploading are final.

Why didn’t one of my starred segments show up while I was riding it?

During an activity, if the GPS signal is poor, it can be hard to determine whether a user is on a particular segment or not. After the activity is uploaded, Strava can examine the entire activity and more accurately determine which segments were triggered - including the ones that didn’t show up in real-time.

Hide/Unhide Segments

Rosie

December 28, 2023

You can hide and unhide segments to customize which segments will be visible to you throughout Strava.

Bad Segment Data

Segments are a community-created/moderated feature on Strava. Anyone can create a segment, which means they will sometimes unknowingly create duplicates or segments from bad GPS data. In addition, they will sometimes unknowingly create segments with incorrect spelling or inaccurate location names. Our support team isn't able to delete duplicate/inaccurate segments or edit their names, but the community can up and downvote segments using the star and hide features. Some examples of bad segment data are:

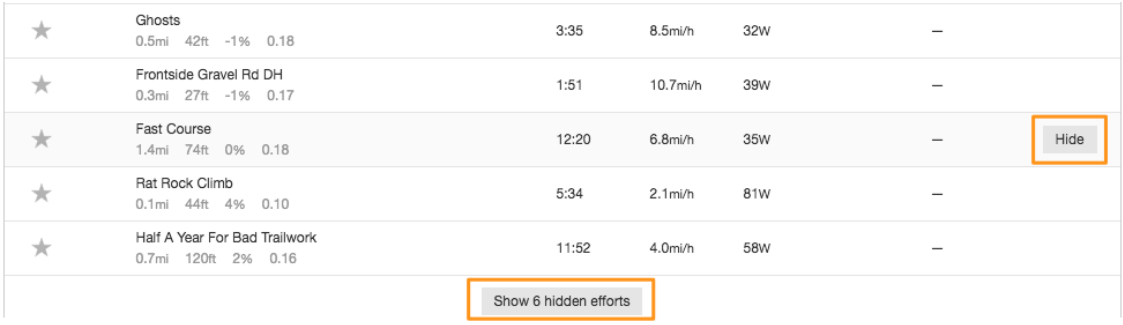
- **Bad elevation data** exists if the activity from which the segment was created had bad elevation data. Here is some more information on how elevation is calculated.
- **Unfavorable start and endpoints** can greatly affect whether a segment will create good matches. Here is some more information on optimizing segment creation.
- Original segment data has **GPS errors, or GPS drift** may lead to inaccurate matches.

Customizing which Segments will be Visible

In segment-dense areas, displaying all segments on one ride is too hard. Depending on the number of times a segment is viewed, that segment is either shown on the ride by default or "hidden" due to a low popularity rating. The user can customize which segments are shown on Activity Detail pages and which are not. By "Hiding" a segment, you vote to decrease its popularity and hide that segment from all of your activities. When you "Unhide" a segment, you increase its popularity rating and vote to show that segment on all of your activities. The overall popularity of a segment will determine whether it will be visible or hidden on your activity pages if you currently don't have preferences set for that segment. You will always be shown segments that you have created, starred, unhidden, or set a goal on.

Hide Segments

- Hover over the segment row in the segments tab at the bottom of an activity
- A "**Hide**" button will appear in the rightmost area of the segment row
- Clicking "**Hide**" will prevent you from seeing this segment listed under any activity throughout the site
- To view all hidden segments for an activity, scroll down to the bottom of the page and click "**Show Hidden Efforts**."



Unhide Segments

- Navigate to the activity page that includes a hidden segment
- Scroll down to the bottom of the page and click "**Show Hidden Efforts**."
- When you mouse over a hidden segment, an "**Unhide**" button will appear in the rightmost area of the segment.
- Clicking "**Unhide**" will allow you to see this segment on all activities throughout the site.

- Please be aware that the segment will not be visible on your activities to other athletes who have not unhidden the same segment. To make the hidden segment visible on your activity, they would need to unhide the segment from their account.

To get Achievements for a Hidden Segment

Segment Achievements are only generated upon upload and ONLY for unhidden segments. Segments marked "Hidden" on your Activity page upon upload will not receive any achievements. To generate the achievement for a hidden segment, refer to the following instructions:

- "Unhide" that segment if it's a cycling segment and star the segment if it's a running segment.
- Change the activity type to something else and then back again to the proper activity type (this triggers segments to be reset.)
- "Refresh Activity Achievements" using the button located in the actions menu (the ellipsis).

Segment Updates: Verified Segments, Decluttering and Leaderboard.

Rochelle Garcia

November 1, 2024

We are renewing our commitment to keeping Segments clean by decluttering duplicate and unused Segments, adding verified Segments, and introducing new rules on Segment leaderboards. With these changes, you can more easily find and compete in the places your community is running, riding, and moving.

Segment Decluttering

We're combing through our Segment database, deleting any duplicate and unused Segments with insufficient GPS data. That way, you can find the Segments you care about faster.

FAQs

How many duplicate Segments are there?
There are 610,000 duplicate Segments around the world that we will remove. Going forward, we'll do more to prevent new duplicate Segments from being created so the feature can remain clutter-free.

I created a goal on a Segment or starred a Segment, which would be duplicates. Will you delete my goal or remove my star?
Your stars and goals will remain. When we remove Segments, we'll only delete duplicates. We'll identify one Segment to keep and move goals and stars from any duplicates to the remaining Segment. You won't lose any functionality related to your Segments if they are identified as duplicates.

What happens if I have a goal or a star on a Segment that Strava has determined to have "bad GPS data?"
If you starred a Segment that Strava makes private, the live Segment functionality will go away. We are doing our best to not make Segments that have been starred private.
Strava is not making Segments that have active goals private unless those Segments have GPS data that won't allow us to maintain an accurate leaderboard. We'll prioritize keeping good Segments that the community is using. If you find a Segment you were using has been deleted, you can always create a new one.

If I have an achievement like a CR or a Top 10 on a deleted Segment, will you delete my achievement?
If you have an achievement on a Segment that has been labeled a duplicate, you should also have a matching achievement on the Segment that will be kept. We have selected duplicate Segments with the goal of preserving your achievements.
If you have an achievement on a Segment that we have determined is irrelevant or has poor GPS data, you may lose that achievement. We want to be sure Segments are fair to compete on, so we need to remove certain Segments with poor data which affect the results. This will only affect a few hundred thousand Segments around the world.

How can I find new Segments to compete on?
On mobile, from the "**Maps**" tab, tap the Map Features selector in the top left-hand corner (this should be auto-selected to **"Routes"**). From there, you can select **"Segments"** and have your map populate with popular Segments in the map area you're viewing.
From the web, visit https://www.strava.com/maps, click the **"Segments"** button, and select any filters you'd like to apply to your search. Your map will populate with popular Segments in the map area you're viewing.
Pro tip: Change your activity type to see Segments specific to your sport type! On web, you'll be able to see Segments for all GPS-based sports.

Verified Segments

The most iconic and popular Segments will now have a badge to signify that the distance and elevation stats have been verified and that Strava (and the community) recognize that section of road or trail as an "official" Segment. So you know that when you're going for the KOM on Hawk Hill, it's THE Hawk Hill. Learn more about verified segments here.

Leaderboard Cleanup

With the introduction of an upgraded auto-flagging system, you can now expect to see 33% fewer impossible efforts on cycling leaderboards. We know 33% isn't 100%, but it will mean many millions of illegitimate efforts will be removed from leaderboards!
We've implemented this system on Segments in the past, but this time the auto-detection will also be gradually applied to older activities. This way, we'll be able to remove past impossible efforts from leaderboards. So in the coming weeks, you may see yourself move up the leaderboards – congrats!

FAQs

Why do I still see incorrect efforts on my favorite leaderboard?
There are billions of activities that need to be checked to catch all the illegitimate ones, so this is going to take a while. We appreciate your patience.

What can I do today if I find an illegitimate effort?
If you see an activity that is inaccurate, you can always flag the activity by viewing it on the desktop. Read more about how to report cheating on Strava here.

Why can Strava only catch 33% of illegitimate segments and not 100%?
It takes a nuanced approach to automatically differentiate between an e-bike ride, a slow-moving car, or a human-powered ride, and we're working toward perfecting that process.

Elevation

Meg

May 31, 2024

What is it?

The elevation profile gives a visual model of the amount and severity of climbing and descending in your activity and where on the route it occurred.

How do I use this?

When set against metrics like pace, cadence, heart rate or power, you can see how elevation and incline (uphill or downhill) affected your output, effort and speed.
This can give you a more complete picture of your activity and whether you met your goals than metrics like pace or speed alone.
For more details, read the extended help article below.

Strava uses a variety of methods to calculate elevation

- How elevation is calculated on segments, activities, and routes
- The details
  - Elevation databases
  - Barometric altimeters
  - Known issues and problematic terrain
  - Elevation challenges
- FAQs

How Elevation is calculated

It is important to remember that elevation data from all sources (barometric altimeters, GPS signals, and databases) is subject to error and that the elevation gain for an activity is an estimate. We are always working on improving our elevation data sets and algorithms.

Elevation on a segment. Segments are always created from a Strava activity and as a result, they inherit some of that activity's characteristics.

- If the original activity was recorded by a GPS device **with a barometric altimeter**, the segment will use the barometric data recorded by that device.
- If the original activity **was not recorded by a barometric device,** or if a Strava admin has rebuilt the elevation for that segment, then we will cross-reference the GPS data to Strava's elevation base map in order to find the elevation.

Elevation on activity pages will be calculated in one of two ways: cross-referencing the GPS data to a database or by using the barometric data in the original file.

- **A device with barometric altimeter:** If the GPS device used to record your activity has a barometric altimeter, Strava will use the elevation data in the original file. We do some smoothing to the data, which includes discarding outliers to reduce noise. The amount of smoothing is **less** on activities with barometric data than it is for activities without barometric data. If your device has a barometric altimeter, but your elevation is being recalculated, please **refer to the FAQs**
- **A device without a barometric altimeter and the Strava mobile app:** If your activity was recorded with the mobile app or with a GPS device that does not have a barometric altimeter, Strava will use your activity's GPS data to "lookup" the elevation for each point you record. We do some smoothing to the data, which includes discarding outliers to reduce noise. The amount of smoothing is **more** on activities without barometric data than it is for activities with barometric data. While we attempt to match the path of your activity with the actual trails and roads in the area, elevation calculation is still somewhat dependent on your GPS data and the accuracy of the database for that region. If you have questions or feedback about your activity's elevation data

please let us know by filling out this form. Please note that your responses may be used to improve the elevation basemap or fix any issues, but you may not hear back from us.

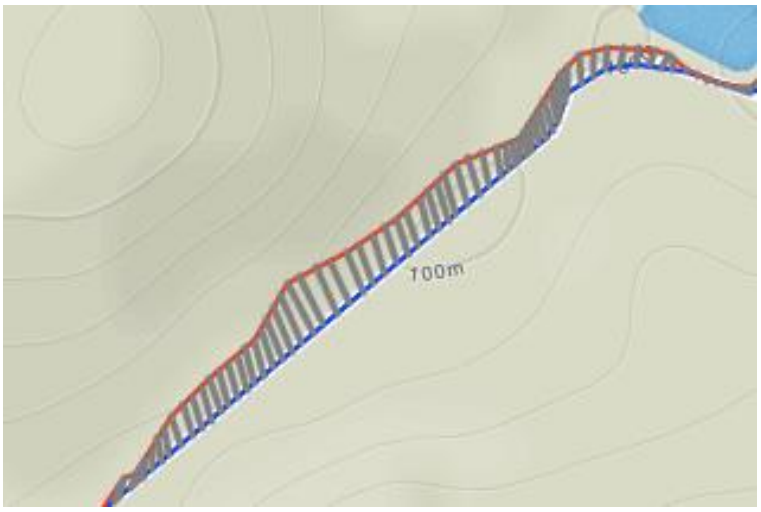
Elevation on a route. Our Strava Route Builder uses our own personal database to calculate elevation data. The GPS data of the route gets cross-referenced to this database when Route elevation is calculated.

The Details

Elevation databases

In the past, we consulted public elevation databases to determine the elevation at each point in the activity. The resolution of these databases varied based on location and we were not able to control for the many errors in these databases. We now have an elevation look-up service powered by data from the Strava community. Each activity that is uploaded from a device with a barometric altimeter is used to build the Strava elevation basemap.

To account for inherent variability in GPS data, we use an algorithm to ensure that we are "looking up" the elevation for the road or trail you were *actually* on. The red line in the image below is an example of GPS data that is drifting off the actual trail (represented in blue). Our algorithm has recognized this and will use the elevation data represented by the blue line instead of the actual, and slightly "off," GPS data.



Limitations of the Strava elevation basemap

There may not be basemap data available for all or part of your activity. In this case, we will use a public database to look up your elevation data. A dense road network or poor quality GPS data could also prevent us from choosing the correct reference points in the basemap or public database.

Barometric altimeters

Barometric altimeters determine altitude by measuring atmospheric pressure. Measurements can be affected by changes in the weather or the sensor holes getting blocked by water or other debris. Device manufacturers may provide basic calibration procedures. For example, Garmin devices with barometric altimeters allow a certain number of manual elevation points to be set; starting an activity near an elevation point causes the device to use the known elevation as the starting point. Consult the manual for your device to learn about calibration procedures. We generally consider the data from barometric altimeters to be of higher quality than the data derived from GPS signals and prefer that when processing activities.

Typical problems with barometric devices
The elevation profile looks like it is drifting over time. This can happen with barometric altimeters as the weather changes (eg. the atmospheric pressure rises or drops, causing the altimeter to measure as though it is at a lower or higher altitude).

Other Known Issues and Problematic Terrain

- *The activity has areas where the elevation is zero or is very linear*. This can happen when crossing bridges or through areas that we don't have coverage for. Coastlines can be particularly problematic. We attempt to fill in the data by interpolating data; the interpolation depends on where the missing data begins and ends.
- *The elevation profile has climbs, but the elevation gain is very small.* We smooth the data before calculating gain and depending on the resulting data, elevation changes may not be enough to pass a threshold that we use for determining whether or not you have gained elevation.
- *The elevation profile has climbs, but the elevation gain is very large*. This often happens with activities recorded in mountainous areas outside of the US. The resolution of the data sets can be coarse and the profile may be bumpier than it should be even with interpolation and smoothing. Sometimes they are so coarse it's like trying to find a smooth elevation profile over a lego block model of your area. These bumps may pass the threshold in gain calculations and the total gain from the bumps may add up to more than the gain that was perceived on the ride.

Elevation Challenges

Strava hosts a variety of athletic challenges for our athletes, some of which have elevation/climbing goals. Due to the high visibility of these challenge leaderboards, elevation-based errors will surface more easily than they would normally. As mentioned above, there are many uncontrollable variables involved in calculating elevation gain. These variables can lead to exaggerated elevation totals, particularly if the activity takes place in an international, mountainous, or coastal region. If you notice activities with inaccurate elevation gain, **please do not flag them**. Most of the time this is a result of poor database resolution, which is out of the athlete's control.

Climb Categorization

Rosie

December 28, 2023

The Strava climb categorization is similar to how the UCI categorizes climbs but with some modifications. When a climb is categorized, for the Tour de France for example, there is a subjective component to the categorization. If a climb is at the end of the stage it could get a tougher category than if it was earlier in the stage. Strava's method is objective, so if a climb is a category 1 climb it will always be a category 1.

Determining Climb Categories

A segment can be categorized as a climb on Strava if the length of the climb (in meters) multiplied by the grade of the climb is greater than 8,000. Please be aware that some segments may be categorized climbs but not marked as such.

You will see that a segment is categorized by the number on both the segment page and on the list of segments on your activity page. The categories are broken down as follows:

- Cat 4 > 8000
- Cat 3 > 16000
- Cat 2 > 32000
- Cat 1 > 64000
- HC (Hors Categorie) > 80000

Route Difficulty

Meg

April 9, 2024

Strava estimates the difficulty levels for hiking, walking, or running routes as **Easy**, **Moderate**, or **Hard**. Mountain biking and gravel ride routes can be categorized as **Easy**, **Moderate**, **Hard**, or **Extreme**. Route difficulty is unavailable for the regular cycling sport type.

We calculate route difficulty based on attributes like distance, elevation, and aggregated community completion time. Routes are assigned an overall difficulty level but you can also view how the difficulty level may vary along the route from the details page.

Overall Difficulty Level

- The overall difficulty level can be seen on the **Maps** tab from the Strava mobile app for both generated community and **Saved** routes. An **Easy**, **Moderate**, **Hard**, or **Extreme** designation can be seen just below the route name.
- On the Strava website click on either **Maps** from the top navigation bar or **Dashboard > My Routes,** to access your saved routes. Click on the route details page to view the difficulty level below the save and edit options.

Difficulty Along the Route (Mobile-Only)

- Tap on any generated community or **Saved** route of the qualifying sports types and scroll down to the "**Terrain**" section. Swipe to the difficulty map to see the route's polyline displayed as a gradient of color based on the difficulty. Easier sections will be green, moderate sections will be blue, and hard sections will be black.

Please be aware that route difficulty is unavailable for the ride sport type.

Best Efforts - Overview

Meg

December 9, 2024

See your personal records (PRs) and performance trends in several running and cycling categories with Best Efforts.

Best Efforts tracks your fastest times at benchmark running and cycling distances, as well as your best power outputs on the bike.

Your performances are tracked automatically with GPS and power data from your running and cycling activities. You can edit an effort or remove it from Best Efforts if needed. For more information and troubleshooting help with Best Efforts, tap here.

Using Best Efforts

On the mobile app, access Best Efforts from the Progress tab, where your **most recent lifetime top three performances** are displayed at the entry point. Tap through to see your lifetime PRs and further analysis in each category.

Lifetime PRs

In the main Best Efforts module, toggle between categories to see all your **lifetime** PRs and second- and third-best performances in each. Click the drop-down on a performance to see the accompanying activity.

View Analysis & Power Curve

Within each category, tap **View Analysis** to see your **top ten lifetime** performances, **top performances by year,** and a graph of your **trend over time** in that category.
Analysis is not available for the cycling power category – instead, click **View Power Curve** to see your best power outputs over time.

Best Performances from Each Activity

On each run and ride you upload, the activity details page includes a Best Efforts section with your fastest times for every benchmark distance and/or power interval you covered in that activity. The activity will also note if you recorded a lifetime top three achievement.

Remove Effort

Select “Remove Effort” under a result to remove the performance from your Best Efforts history and erase the achievement banner from the activity. **This will not delete the activity.** If you remove a PR, your next-fastest time will become your PR.

Edit Effort – Running Only

If you know a running performance is inaccurate, you can correct it. Click the **“Edit Time”** box under the result to enter a precise chip time from a race or correct for faulty GPS data.
Changes will be reflected in Best Efforts, PR Analysis and the Results module of the activity. (This might cause the achievement banner to be removed from the activity.) The GPS data of the activity cannot be altered.

Power

Meg

May 31, 2024

https://support.strava.com/hc/en-us/articles/216918457-Power

What is it?

Power is the amount of energy a cyclist or runner is generating, measured in watts.

Your power as a percentage of functional threshold power (FTP), or the number of watts you can hold for one hour, is a key indicator of effort level and helps determine the zone or zones you worked out in. (You can change or enter your FTP in profile settings.) A power meter or smart trainer is required to track this for cycling. For running, it's estimated based on your weight and pace.

How do I use this?

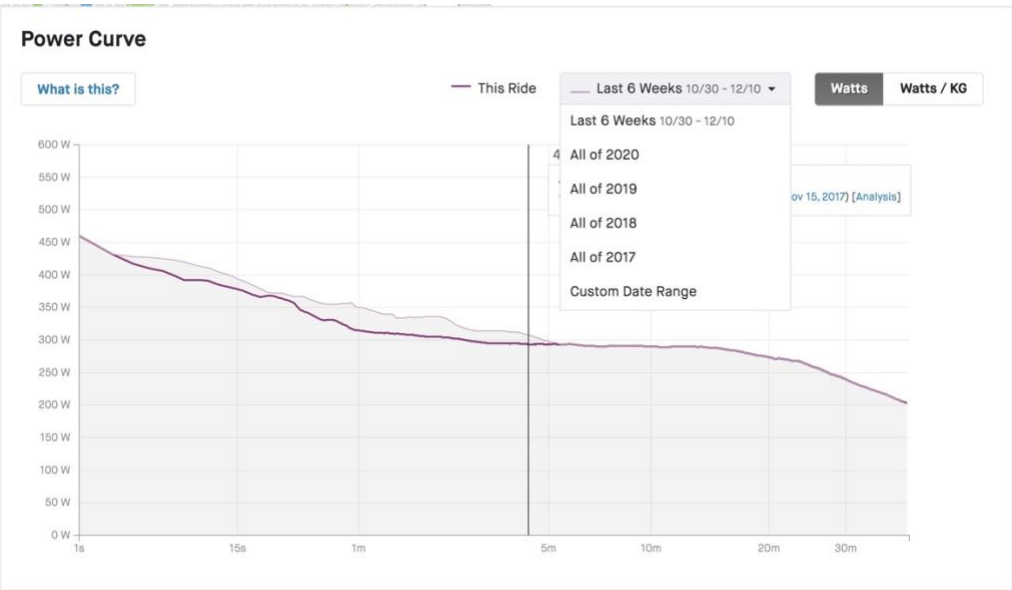
Overall, your power gives you a good read on how strong you were – the higher the number, and the easier it felt, the better.

You can also see the impact this activity had on your training with Power Zones: Z1-2 is easy and ideal for recovery, Z2-3 is good for moderate endurance, and Z4 and up is considered intense. For cyclists, your Power Curve shows the maximum power you held for various time intervals during your ride (e.g. 5 seconds, 5 minutes), and you can compare it to previous Power Curves to see if you're improving.
Read the extended help article below for more details, including determining your FTP.

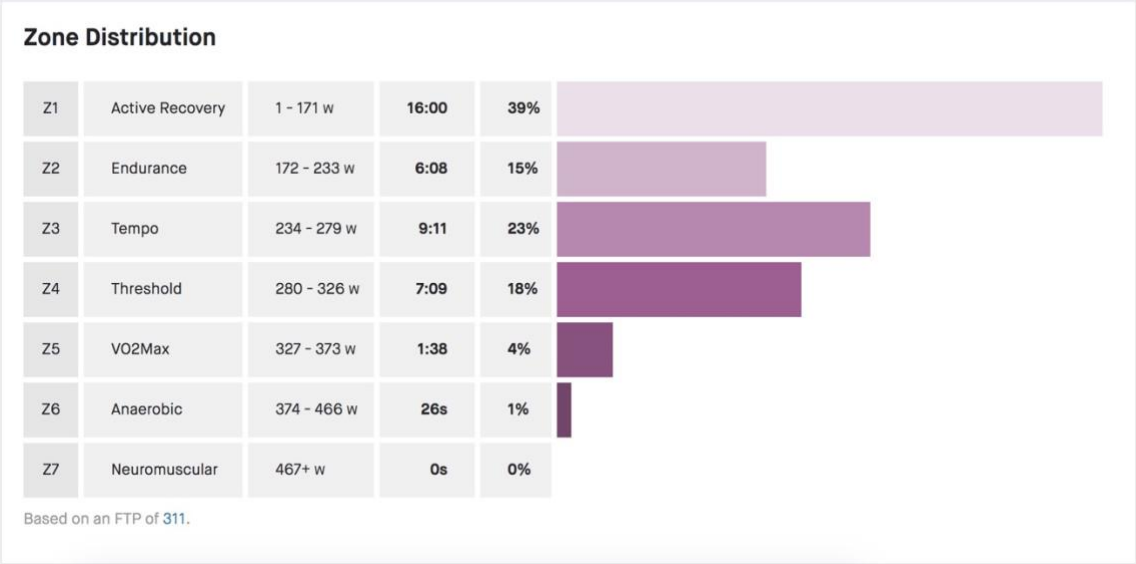
Strava subscribers who ride with power meters can start training more intelligently with advanced power training analysis.

Available Features:

- **Weighted Average Power** looks at all of your power variations and provides an average power for your ride. This is a better indicator of your overall effort than simply taking your average power.
- **Training Load** is calculated by comparing your power during a ride with your Functional Threshold Power (FTP) and seeing how much load you put on your body during the workout. Training Load is a great way to determine how much rest you need after your workouts.
- **Intensity** is Strava's way of showing how difficult a ride was. Strava looks at your Weighted Average Power for the ride and compares it to your FTP. For example, if your Weighted Average Power is 225W and your FTP is 300W, then your Intensity would be at 75%.
- **Power Curve** shows your best average power for time periods of one second up to the length of your ride. We pinpoint your best efforts during a ride and you can compare them with your best efforts in the last six weeks, the current year, and years past. The Power Curve can be displayed in Watts (W) or Watts per Kilogram (W/kg.) Other athletes may be able to deduce weight using information displayed in the power curve. To control which stats are visible on your activities, see our article here.



Power Distribution by Power Zones: While the Power Curve shows your best efforts for given periods of time, Power Zone charts take each second of power from your ride and distribute it by training zone based on your FTP.

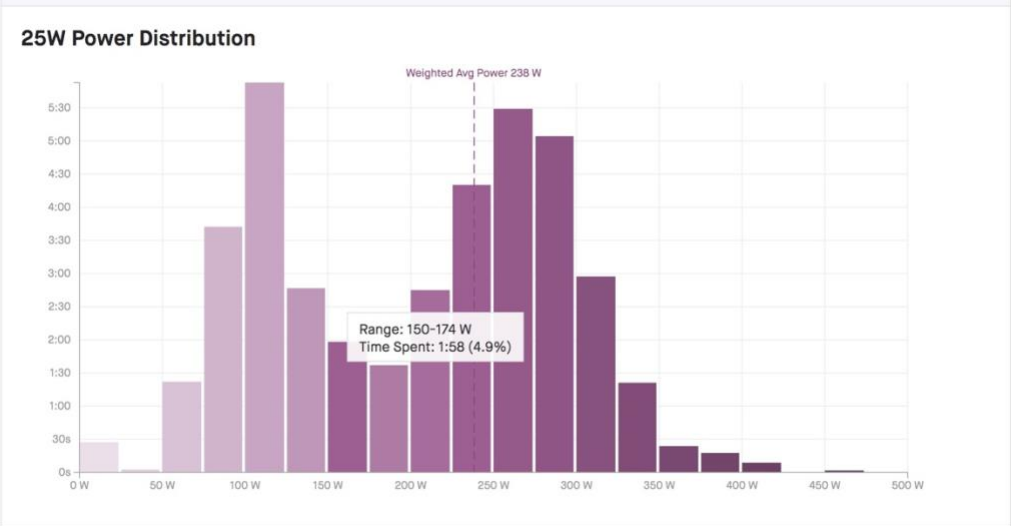


Zone Distribution

Z1	Active Recovery	1 - 171 w	16:00	39%
Z2	Endurance	172 - 233 w	6:08	15%
Z3	Tempo	234 - 279 w	9:11	23%
Z4	Threshold	280 - 326 w	7:09	18%
Z5	VO2Max	327 - 373 w	1:38	4%
Z6	Anaerobic	374 - 466 w	26s	1%
Z7	Neuromuscular	467+ w	0s	0%

Based on an FTP of 311.

Power Distribution by 25W Increments: Similar to the Power Distribution by Power Zones, the Power Distribution by 25W Increments simply puts each one second of power into a 25W training zone from 0 to the highest power output on the ride.



Power Curve in Training Section: Similar to the Power Curves on your activities, the Power Curve in the Training section allows you to compare any two of your power curves. For example, you can compare the last six weeks to all of the previous years and see how you're progressing.

Calculating Your FTP

We recommend you test for your FTP at least every few weeks to a month while you're training. Here are some tips to get the most out of your FTP testing:
It's extremely taxing on your body (and your training program) to continuously push out 60-minute max efforts. It's also difficult to find a stretch of road where you can ride for 60 minutes uninterrupted and maintain a steady wattage. Thus, the easiest way to calculate your FTP is to test your best average power for 20 minutes and then subtract 5%. We believe 20 minutes is enough time to stress the same physiological systems as a 60-minute effort would and it is easier to consistently do within your season.
Try to reproduce the same conditions for each test, for example, the same stretch of road or the same trainer/rollers.
Make sure you are fresh (the previous few days should be light in terms of training load).
Properly warm-up.

FTP Management in Settings
On the web, go to your Settings page by hovering over your profile picture in the top right and selecting **Settings**.
Click on the **My Performance** tab on the left side of the page.

i) **Click to edit** under Functional Threshold Power towards the bottom of the page.
ii) You can also add date ranges for your FTP in your **My Performance** settings on the web. Old rides will retain the zones and stats for your fitness at the time instead of updating to match your current output. You'll also see your past numbers reflected in your Fitness graph.

On the mobile app, select **You** > **Profile** and select the option to **Edit Profile**.
Enter your **Functional Threshold Power** in the appropriate field under Performance Potential.

Power Curve - Cycling

Meg

January 2, 2024

See and compare your best power outputs with the Best Efforts power curve. Power data from your rides collected via a power meter or smart trainer is required to produce your power curve.

The power curve shows your **best average power for time periods of 1 second up to the length of your longest ride** from the time period shown.

You can display and compare **curves from multiple time periods**. Select the time period(s) at the top of the power curve page, where you can choose to display the power curve from an entire year, the last six weeks or a custom date range. (Custom date ranges must consist of whole weeks – if you select a mid-week start or end date, it will adjust to the prior Monday or the following Sunday, respectively.)

While the power curve shows your best average power for **every time interval** you've covered, the Best Efforts landing page **only tracks your power PRs and second- and third-best performances** over 5 seconds, 30 seconds, 1 minute, 5 minutes, 10 minutes, 20 minutes, 30 minutes, 1 hour and 2 hours.
For details on the other cycling categories tracked in Best Efforts, tap here.

Your performances are tracked automatically with power data from your cycling activities. A power meter or smart trainer is required to collect power data. You can hide a cycling effort from Best Efforts if needed.

How Distance is Calculated

Meg

August 22, 2023

Distance is the most basic training statistic - therefore, being confident in your distance data is important. There are a few ways to gather distance data while recording a GPS-based activity

- How Strava measures and displays distance
- Methods of Calculating Distance
- Reverting Distance
- Recording distance on a GPS device

How Strava measures and displays distance

When a GPS file is uploaded, Strava takes the distance data recorded in the file and parses it into a data stream to calculate total distance, average speed, and max speed. Depending on which method for recording distance is used (see explanation below), that data will be reflected in the distance stream and thus on Strava. Under normal conditions, differences should be minimal when comparing distance or speed metrics on Strava versus the GPS device. Still, any minor inconsistencies are likely due to number crunching on both ends - Strava processes and analyzes the data in the file independently. In contrast, most GPS devices tabulate these values on the device itself.

On Strava, distance contributes to your overall distance totals, whether in your Training Calendar, the bar graph on the Profile page, or your overall and yearly stats in the Profile sidebar. Additionally, distance readings contribute to your average speed, as average speed is calculated from a distance over your total moving time. Distance does **not**, however, contribute to your segments or segment times. Segment times are based on when you cross the start and endpoints of a segment. Therefore, distance is mostly a personal metric and statistic, except when Strava runs a distance-based challenge, like the monthly distance challenges, and in that case, the distance would be competitive.

Methods of Calculating Distance

There are two main ways to calculate the distance for most sports - **Ground Speed Distance** and **GPS-calculated Distance**. Ground speed will measure your speed along the surface you are traveling (counting the revolutions of a wheel), and GPS-calculated distance will "connect the dots" between your GPS points and triangulate the distance between the coordinates. Each method of gathering data can and may introduce some inaccuracy.

- **GPS-based device approach**: The Strava mobile apps and many GPS devices will calculate your distance accumulated in "real-time" while the device is recording based on the GPS data.
  - **Pros**: Refined calculation to gather distance data built into the file in the distance stream, measured in meters.
  - **Cons**: The complicated nature of this "real-time" calculation can lead to stuck points, where no additional distance is recorded from the previous point, which can cause some Strava calculations like Estimated Best Efforts for Run to fail. Since this is a GPS-calculated distance, a flat surface is assumed, and vertical speed from topography is not accounted for. Also, some accumulated distance may be lost as straight lines connect each GPS coordinate instead of an arc. This calculation method does not capture variations in the route between GPS points and may vary further when battery-saving features are enabled.

- **GPS-based, Strava post-upload approach**: After GPS data is recorded and uploaded to Strava, it is parsed into streams of data and analyzed. At this time, a calculation can be run on the GPS coordinates to get the distance. This is how Strava determines the length of any uploaded file that

does not include a distance stream. You can use this method if you suspect a problem with your device's recorded distance (read about reverting distance down below.)
    - **Pros**: Post-upload GPS-based distance can eliminate problems like stuck points (see above) and create smoother, more accurate distance data than the device equivalent.
    - **Cons**: A flat surface is assumed, and vertical speed from topography is not accounted for. Similar to the above, straight lines connect the GPS points.
- **Speed/Cadence Sensor Garmin GSC-10 approach**: Ground Speed distance is measured by counting the wheel revolutions and then multiplying by the wheel circumference.
    - **Pros**: A wheel sensor will capture vertical speed and the additional percentage of distance accumulated with elevation changes. This could become a slightly more significant factor for mountain bikers who gain and lose a lot of elevation gain rapidly.
    - **Cons**: Common problems with relying on a wheel sensor include: wheel size is not documented accurately, the device is moved to another bike with a different wheel size and not adjusted, the auto wheel size is calculated wrong either because of GPS inaccuracies or because the magnet did not count every wheel revolution.

Reverting your distance

If you suspect there is a problem with the distance recorded by your device, you have the option to override your device's distance with the Strava post-upload approach. Click on the three-dot action menu in the left sidebar and select the option "**Correct Distance.**" This can improve the quality of uploaded data by eliminating outlier GPS data like inaccurate GPS points and data that is inconsistent with the file. If you change your mind, click the button again to revert back to the original distance.

Hierarchy of Garmin device inputs when multiple distance data sources exist

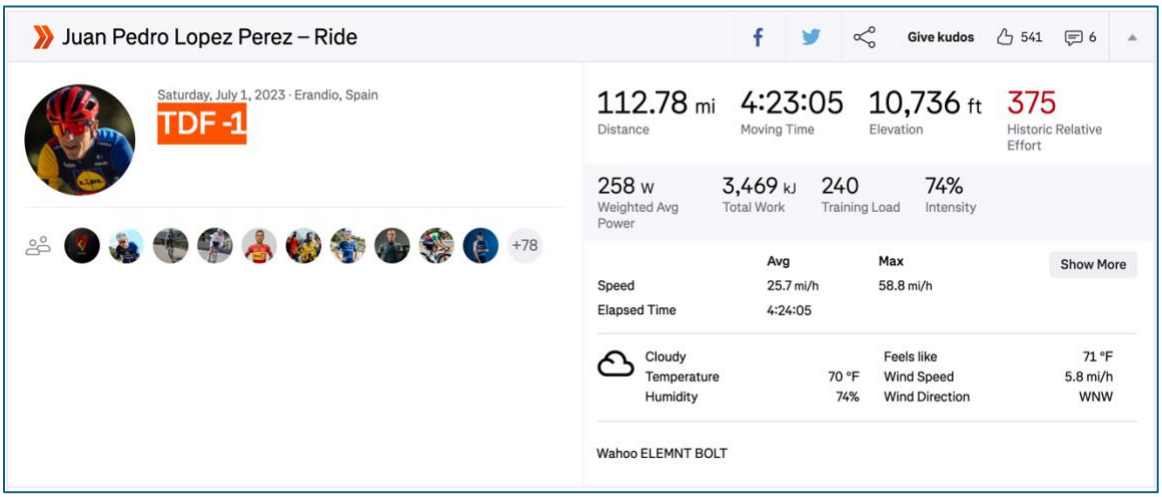
What happens if you have a PowerTap or a GSC-10 Speed/Cadence sensor or both? When the Edge has multiple sources for the same information, it uses a predetermined selection process to go with what it will consider to be the most accurate source.

- If you have a PowerTap hub connected to your Gamin, it will take speed readings from the PowerTap hub above all other inputs.
- If you have a GSC-10 Speed/Cadence sensor, the Garmin will take readings from this output over the GPS-calculated distance.
- If you have neither, Garmin will calculate the distance based on GPS.

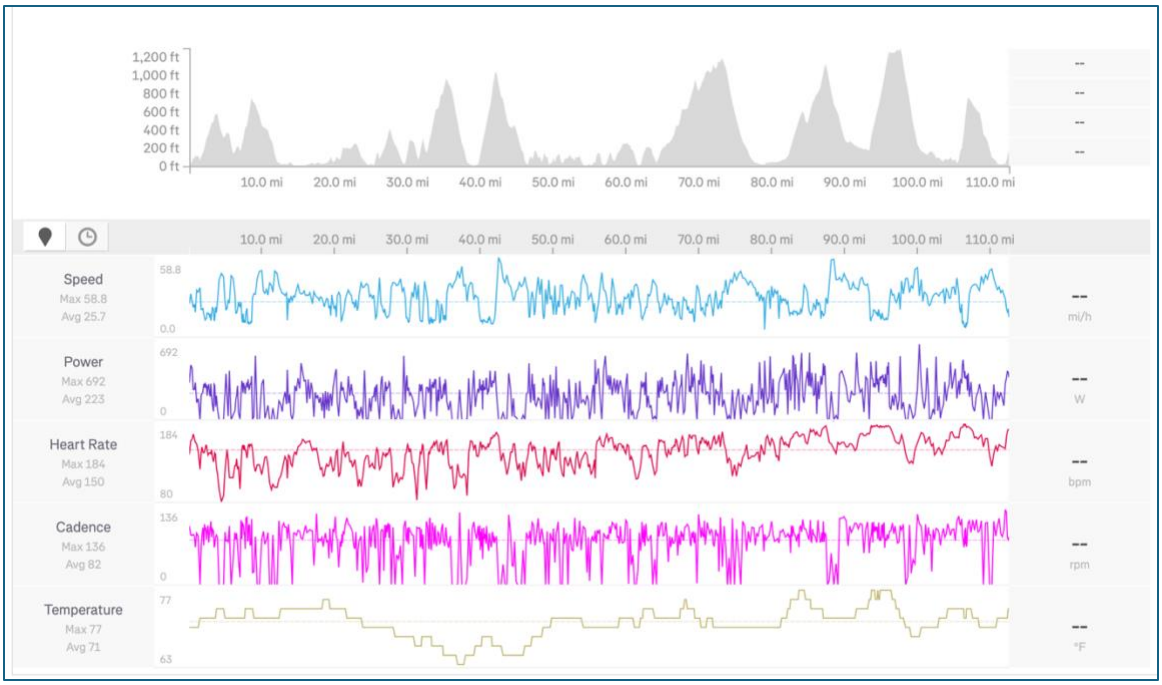
The key is that the data from either source is seamlessly incorporated into the recorded file under the distance stream. In some cases, the speed in MPH is documented in the file also as an extension. Regardless, each Garmin-produced file has a distance stream of data measured in accumulated meters that serves to measure total distance and speed (both max and average).

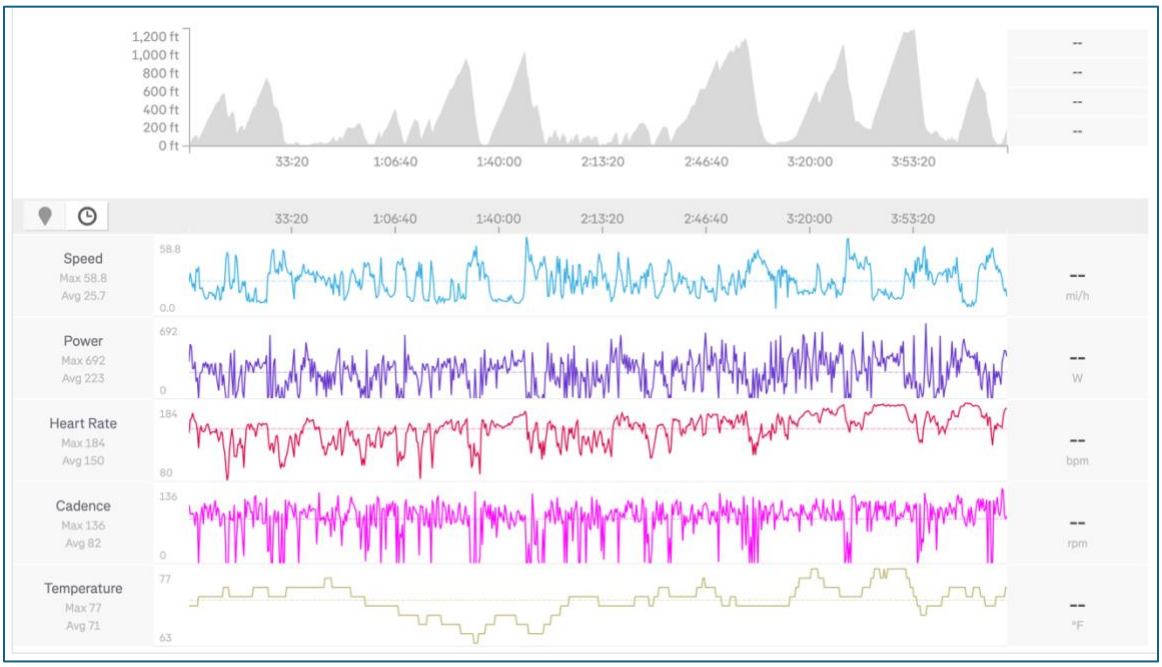
Example Activity- 9368941696

Juan Pedro Lopez Perez – Ride
Saturday, July 1, 2023 Erandio, Spain
TDF -1

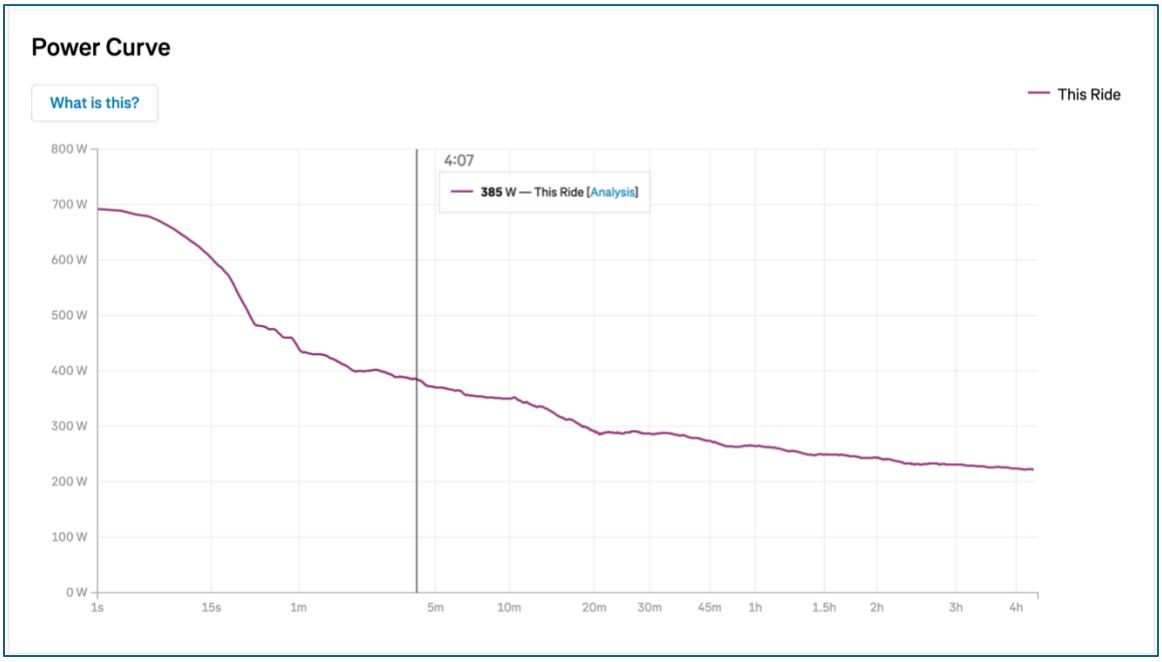


Overview Tab





Analysis Tab



Juan Pedro Lopez Perez – Ride

Give kudos 541 6

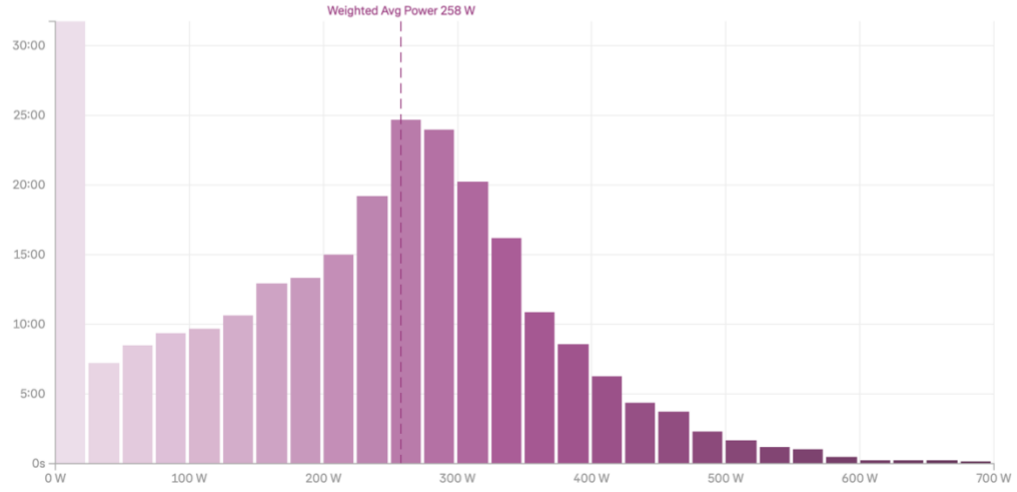
112.7 mi	4:23:05	10,736 ft	3,469 kJ	258 W	240	74 %
Distance	Time	Elevation	Total Work	Weighted Avg Power	Training Load	Intensity

Zone Distribution

Z1	Active Recovery	0 - 192 w	1:39:44	38%
Z2	Endurance	193 - 262 w	49:36	19%
Z3	Tempo	263 - 315 w	50:01	19%
Z4	Threshold	315 - 367 w	31:10	12%
Z5	VO2Max	368 - 420 w	16:34	6%
Z6	Anaerobic	420 - 525 w	12:44	5%
Z7	Neuromuscular	525+ w	3:20	1%

Juan Pedro Lopez Perez – Ride

Give kudos 541 6

112.7 mi	4:23:05	10,736 ft	3,469 kJ	258 W	240	74 %
Distance	Time	Elevation	Total Work	Weighted Avg Power	Training Load	Intensity

25W Power Distribution